NOW POP

MICRODUCK

硬件装配手册

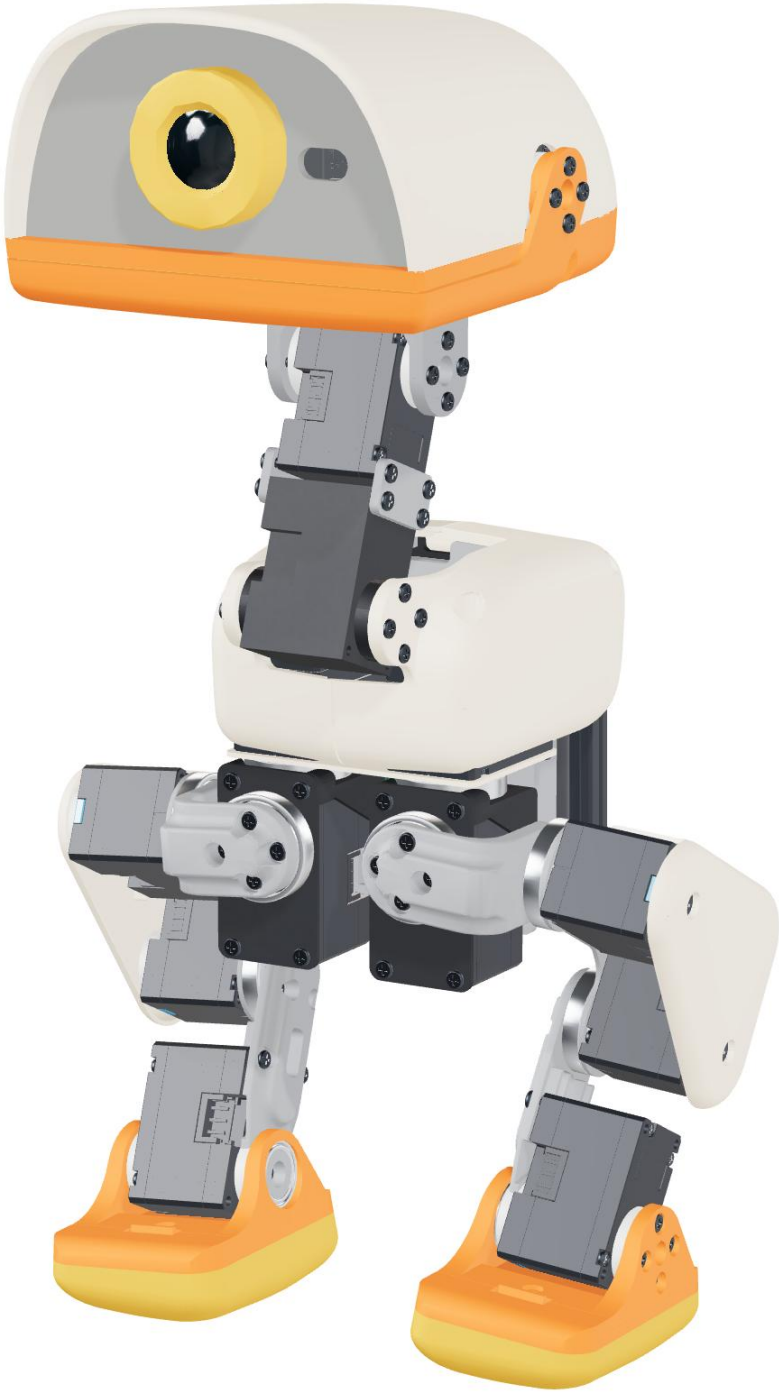
HARDWARE ASSEMBLY MANUAL

制作机器鸭的过程没有伤害任何鸭子。---- Pollen官方

No duck was harmed in the making of this robot. Several were consulted.

V1.0 · 2026-09-25

扫码体验
可视化 Agent 服务
Scan to try the Microduck visual agent



使用前须知 Before you begin

安全 SAFETY

装配全程断电。舵机、主控和电池插拔前先断开电源；电池充电按厂家规格执行。

Keep everything unpowered while assembling. Disconnect power before plugging servos, boards or the battery.

怎么读这本手册 HOW TO READ

每页一步。灰色填充、黑色轮廓的是本步新装的零件；白色细线的是已经装好的部分。黑色胶囊是零件，浅灰胶囊是螺丝的规格与数量。右上角小图标出当前章节在整机中的位置。

One step per page. Grey-filled parts with dark outlines are new in this step; thin white parts are already installed. Black pills name parts, grey pills give screw size and count.

先配软件，再装机械 CONFIGURE FIRST

15 颗舵机必须在装入前逐颗配置 ID 与 1 Mbps 波特率，固件 ≥ v46（《装配与调试手册 V1.2》S1.1–S1.5）。装进结构后再改 ID 会很麻烦。

Give every servo its ID and 1 Mbps baud rate before it goes into the structure (software manual V1.2, S1.1–S1.5).

数据来源 SOURCE

所有图像直接取自 Nowpop 中的 Microduck 模型（Pop3D-ZVP4B）当前发布版本，共 77 个零件、146 个紧固件；螺丝逐颗按它接触的零件归入步骤。

Every drawing is rendered from the live Microduck model in Nowpop (77 parts, 146 fasteners).

供电冲突 VOLTAGE

零售 XL330 额定 3.7–6.0 V，2S 电池满电约 8.2 V。供电方案闭合前，不要用电池给整机带载。

Retail XL330 is rated 3.7–6.0 V; a full 2S pack is about 8.2 V. Resolve the power plan before loading the robot from the battery.

位置示意 APPROXIMATE

该模型是可视化装配，不是官方制造 CAD；螺丝位置按孔位与照片推定，长度以实物孔深为准。头部舵机按位置标注，ID 30–34 装机后按软件手册 S9.2 逐个核对。

A visual assembly, not official manufacturing CAD. Screw positions are inferred; confirm lengths against the real holes. Head servo IDs are confirmed after assembly.

反馈问题 REPORTING ISSUES

发现手册错误或有改进建议，请在下方 GitHub 仓库提交 Issue，写明页码与简短描述，附上标注过的截图更好。我们会根据反馈定期更新手册。

Found a mistake or have a suggestion? Open an issue on the GitHub repository below with the page number and a short description; annotated screenshots are welcome.

这只是一份参考 THIS IS JUST A REFERENCE

组装 Microduck 并不简单。遇到看不清的地方，打开 Nowpop 中的 3D 模型对照，或扫码让可视化 Agent 逐步讲解——整台机器摆在眼前时往往更容易理解。

Building a Microduck can be involved. When a step is unclear, open the 3D model in Nowpop or scan the code and let the visual agent walk you through it.

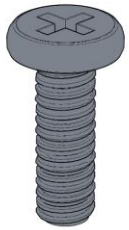
GitHub https://github.com/SIJEYUSES/MicroDuck-Agent/tree/main

NOW POP https://nowpop.art/cad/v/Pop3D-ZVP4B.g3d

目录 Contents

零件参考 Hardware reference 05
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4 电源与电池 Power & battery 37
5 颈部与头部组件 Neck & head assembly 43
6 侧壳与整机合装 Shells & final assembly 56
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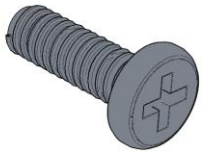
零件参考 Hardware reference



M2×4 十字盘头螺丝

M2×4 pan head screw

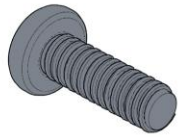
头壳下部等薄壁位置。



M2×6 十字盘头螺丝

M2×6 pan head screw

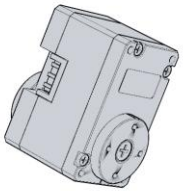
最常用：舵机与支架、连接件。



M2×8 十字盘头螺丝

M2×8 pan head screw

穿过较厚结构件，如小腿、侧壳。



XL330-M288-T 舵机

XL330-M288-T servo

15 颗；装入前完成 ID 与固件配置。



轴承 22×16×4

Bearing 22×16×4

髋、大腿与颈部的回转支承。



小轴承

Small bearing

踝座与下颌的回转支承。

分章零件清单 Parts by chapter

章节	零件	紧固件
1 左腿 Left leg	左脚底、左脚掌、左踝座、小轴承、左踝舵机 · ID 24、左小腿、左膝舵机 · ID 23、左大腿、轴承 22×16×4 ×3、左髋俯仰舵机 · ID 22、左大腿加强板、左髋连接件、左髋横滚舵机 · ID 21、左偏航-横滚连接件、左横滚轴承座、左髋偏航舵机 · ID 20	M2×6 ×30 M2×8 ×6
2 右腿 Right leg	右脚底、右脚掌、右踝座、小轴承、右踝舵机 · ID 14、右小腿、右膝舵机 · ID 13、右大腿、轴承 22×16×4 ×3、右髋俯仰舵机 · ID 12、右大腿加强板、右髋连接件、右髋横滚舵机 · ID 11、右偏航-横滚连接件、右横滚轴承座、右髋偏航舵机 · ID 10	M2×6 ×25 M2×8 ×6
3 躯干与双腿合装 Trunk & legs	躯干底座、轴承 22×16×4 ×2	M2×6 ×8
4 电源与电池 Power & battery	电源支架、IMU 总线板 imu_to_dxl、电池座、电源开关、PCB 卡扣、NP-F550 电池	M2×6 ×12
5 颈部与头部组件 Neck & head assembly	颈座 ×2、颈部舵机 A、颈部舵机 B、颈俯仰件、轴承 22×16×4 ×3、头部偏航-横滚件、头部偏航-横滚舵机、头部电机支架、头部舵机、鸭嘴舵机、下颌、软下颌、小轴承、头壳（下）、面板、M12 镜头座、镜头、眼部饰件、ToF 深度传感器、Radxa Zero 3W 主控、microSD 卡、Robot HAT 载板、扬声器、NFC 天线 · 头部、NFC 天线 · 鸭嘴、软上喙、头壳（上）	M2 ×8 M2×4 ×2 M2×6 ×27 M2×8 ×11
6 侧壳与整机合装 Shells & final assembly	左侧壳、右侧壳	M2×6 ×2 M2×8 ×9
合计 Total	**77 个零件 • 146 个紧固件**	**M2 ×8** **M2×4 ×2** **M2×6 ×104** **M2×8 ×32**

01

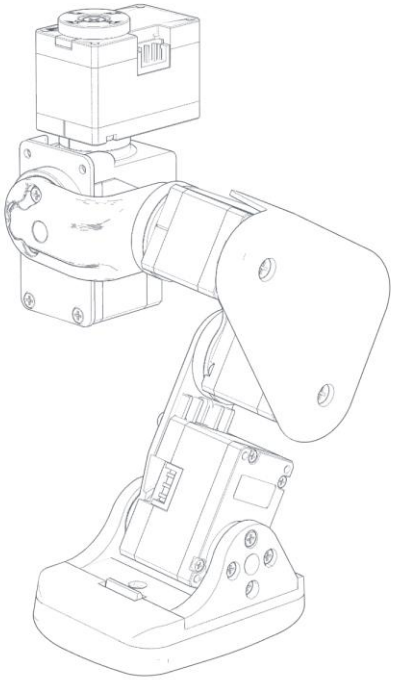


左腿

LEFT LEG

12 步 · L01–L12

左脚底 Left sole	**×1**
左脚掌 Left foot	**×1**
左踝座 Left ankle bracket	**×1**
小轴承 Small bearing	**×1**
左踝舵机 · ID 24 L ankle servo · ID 24	**×1**
左小腿 Left lower leg	**×1**
左膝舵机 · ID 23 L knee servo · ID 23	**×1**
左大腿 Left thigh	**×1**
轴承 22×16×4 Bearing 22×16×4	**×3**
左髋俯仰舵机 · ID 22 L hip pitch servo · ID 22	**×1**
左大腿加强板 Left thigh stiffener	**×1**
左髋连接件 Left hip bracket	**×1**
左髋横滚舵机 · ID 21 L hip roll servo · ID 21	**×1**
左偏航-横滚连接件 Left yaw-to-roll bracket	**×1**
左横滚轴承座 Left roll bearing housing	**×1**
左髋偏航舵机 · ID 20 L hip yaw servo · ID 20	**×1**
M2×6 十字盘头螺丝 M2×6 pan head screw	**×30**
M2×8 十字盘头螺丝 M2×8 pan head screw	**×6**

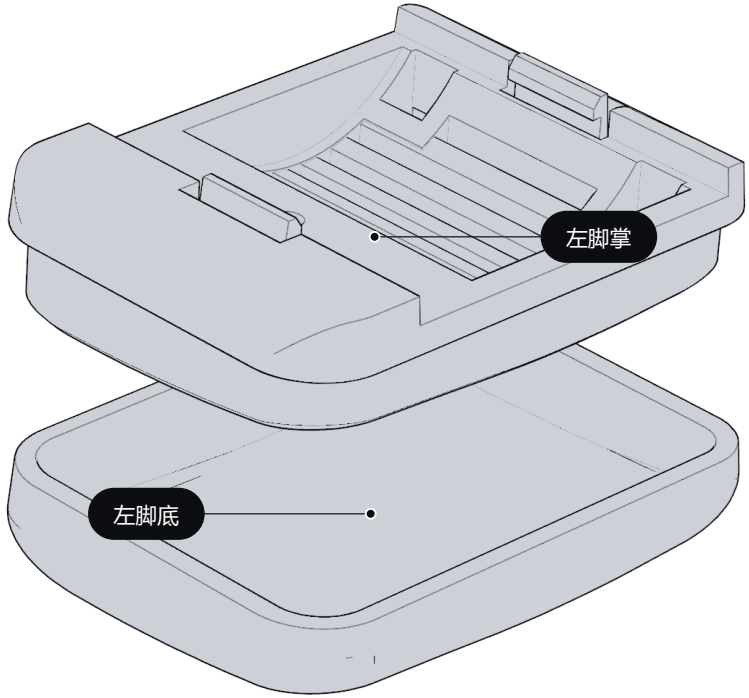


扫码体验 Microduck 可视化 Agent 服务

Scan to try the Microduck visual agent

L01 左脚掌与脚底
LEFT FOOT AND SOLE





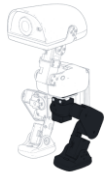
配合方式 FIT

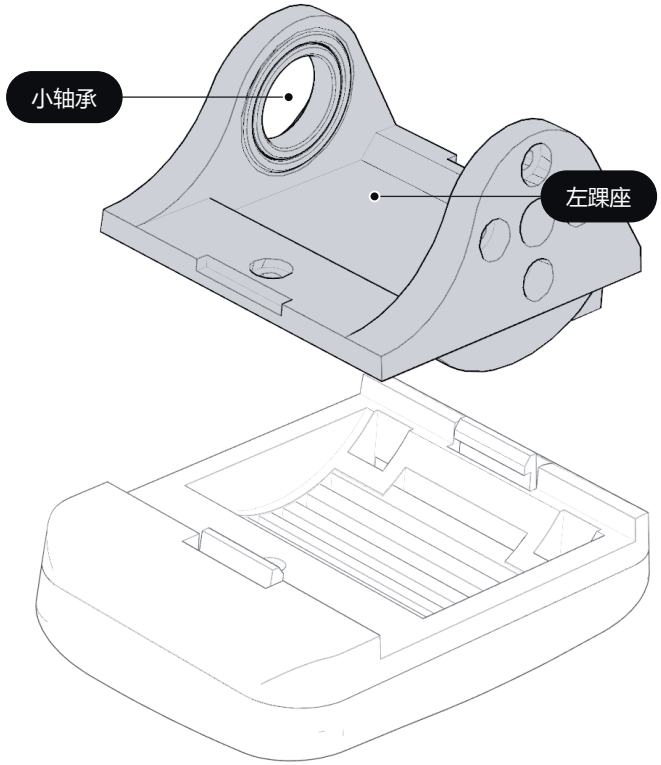
模型中脚底与脚掌之间没有紧固件，按实物卡槽/胶合方式装配。

No fastener is modelled between sole and foot; follow the physical snap or glue fit.

L02 左踝座与小轴承

LEFT ANKLE BRACKET AND BEARING



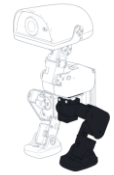


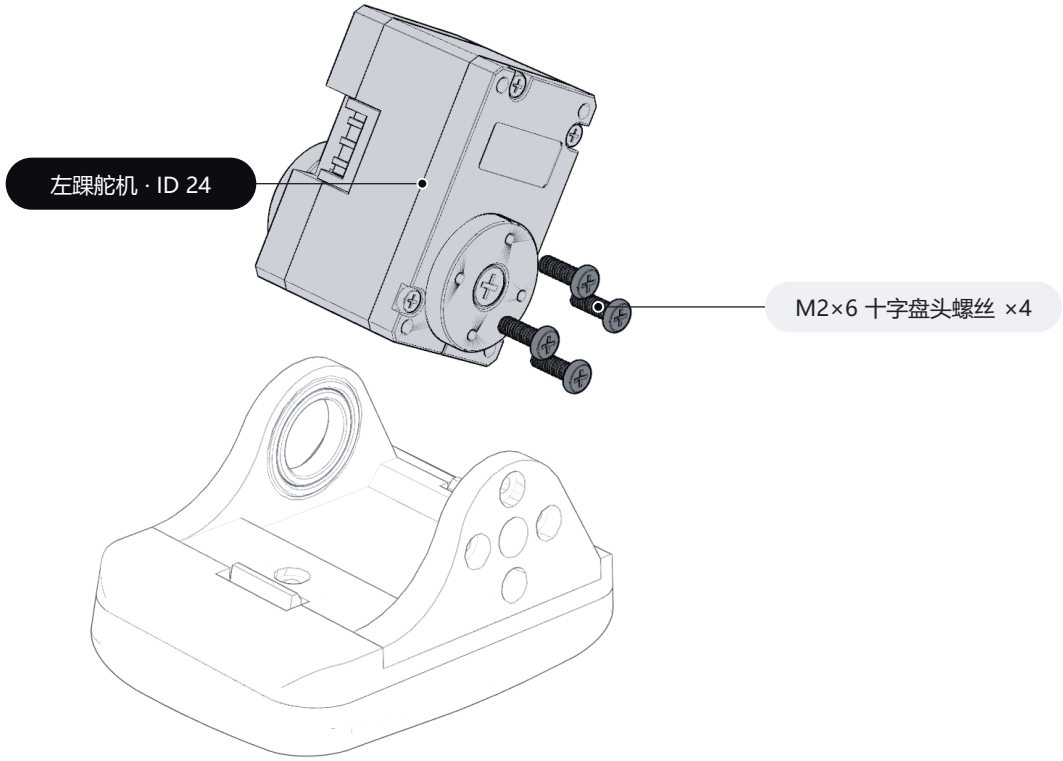
轴承到底 SEAT THE BEARING

小轴承压入踝座轴承孔，端面与孔口齐平。

Press the small bearing fully into the ankle bracket bore, flush with the face.

L03 左踝舵机
LEFT ANKLE SERVO





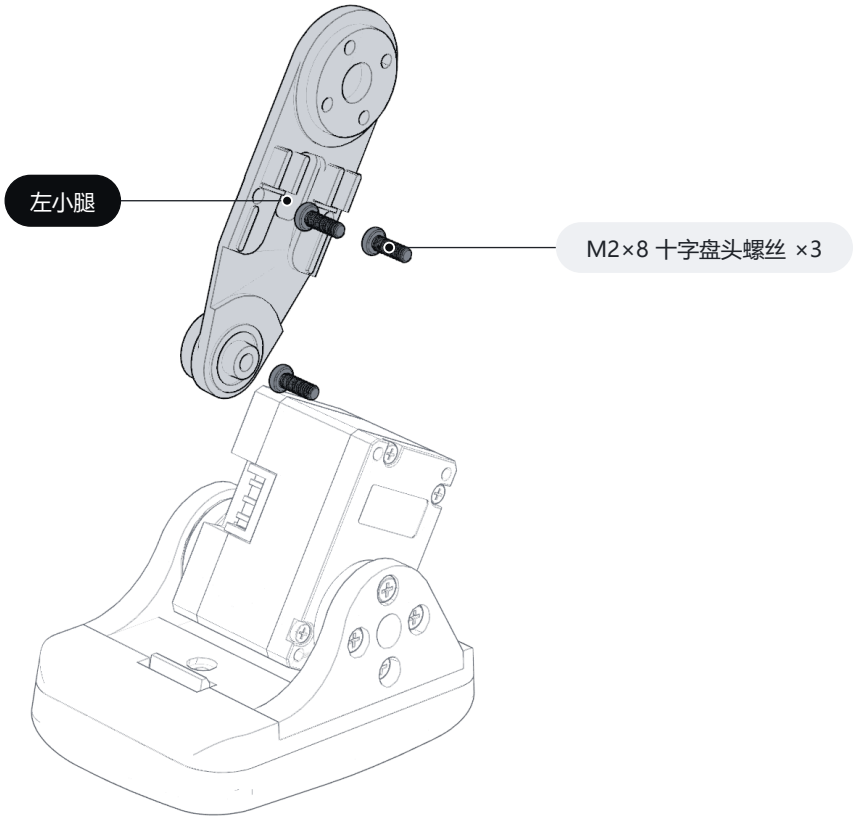
先对零位 ZERO FIRST

舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

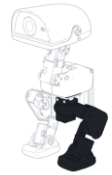
Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

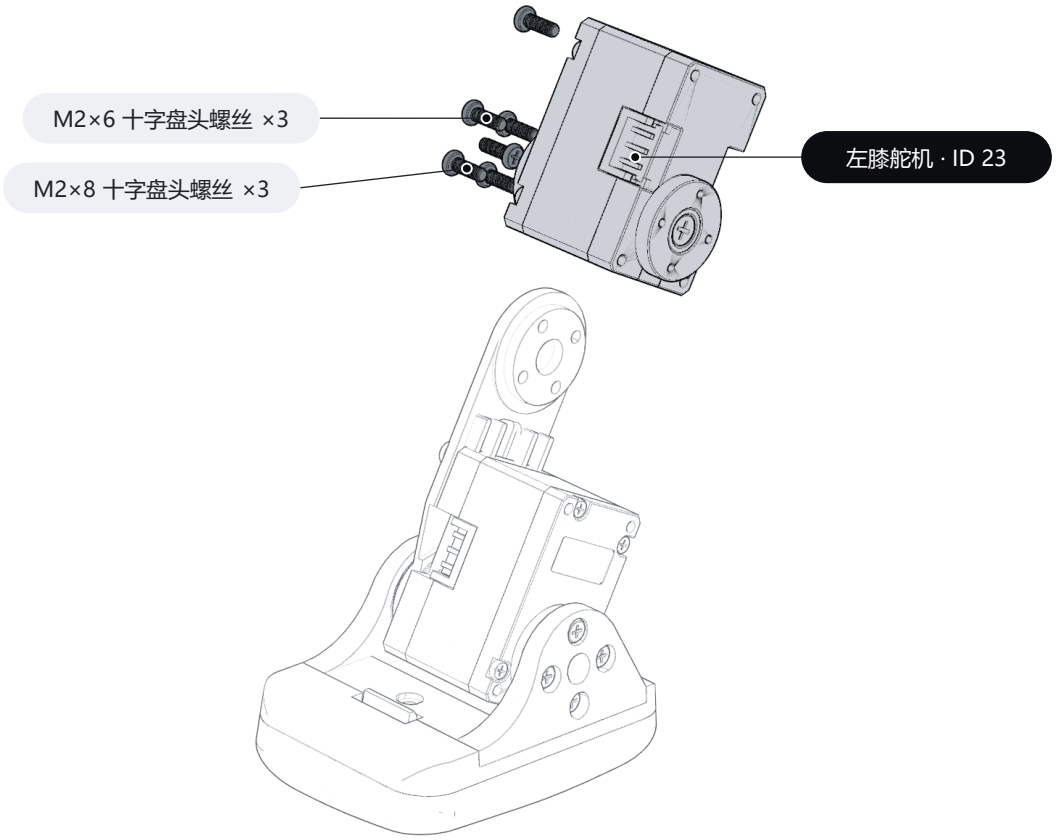
L04 左小腿
LEFT LOWER LEG





L05 左膝舵机
LEFT KNEE SERVO





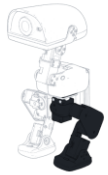
先对零位 ZERO FIRST

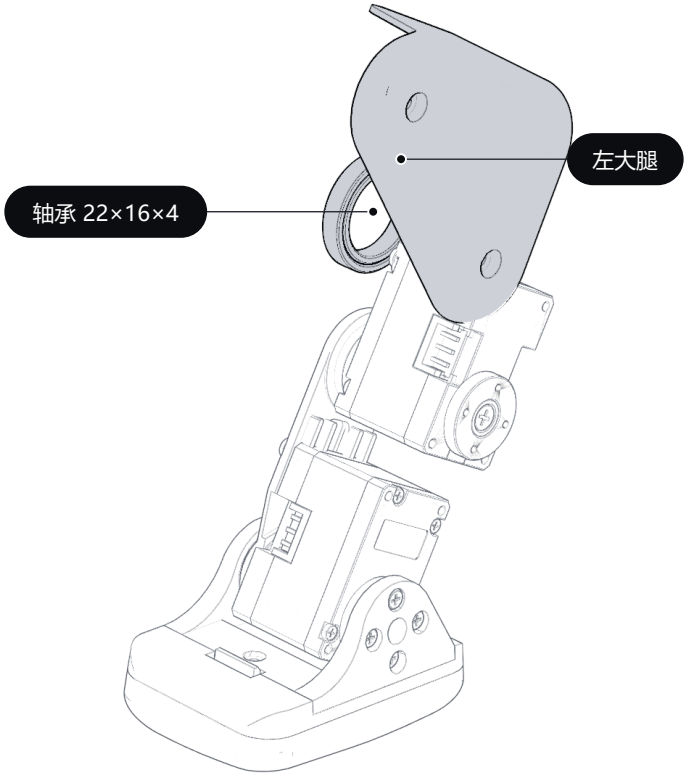
舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

L06 左大腿与轴承

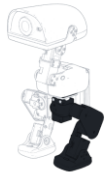
LEFT THIGH AND BEARING

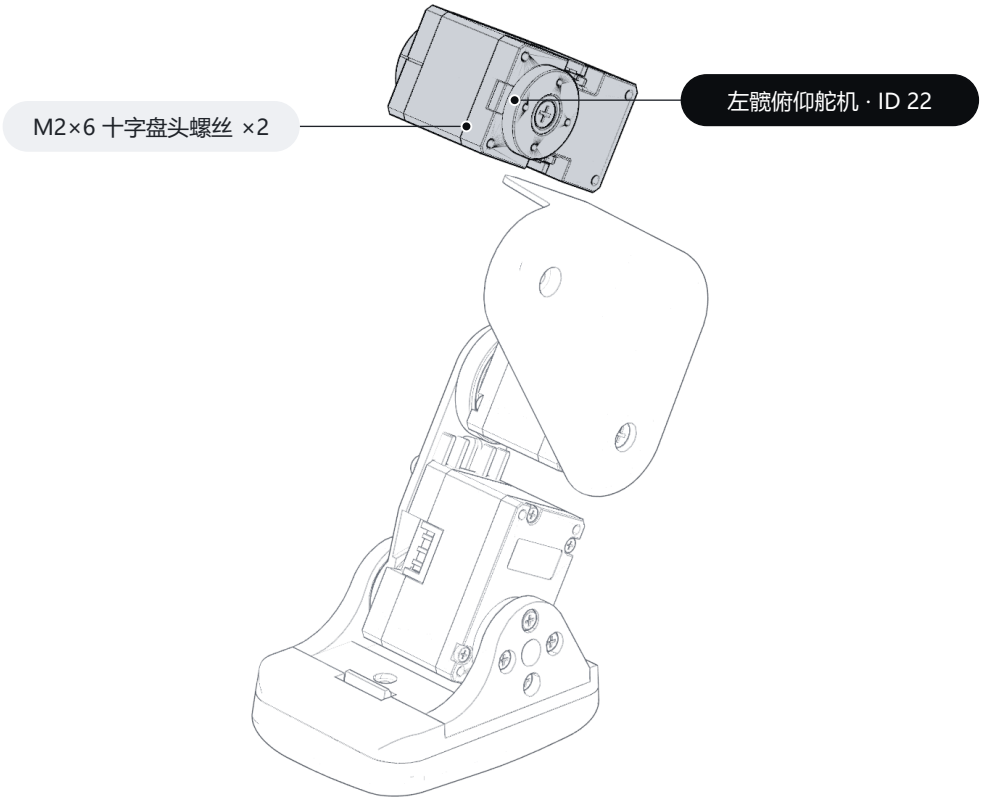




L07 左髋俯仰舵机

LEFT HIP PITCH SERVO





先对零位 ZERO FIRST

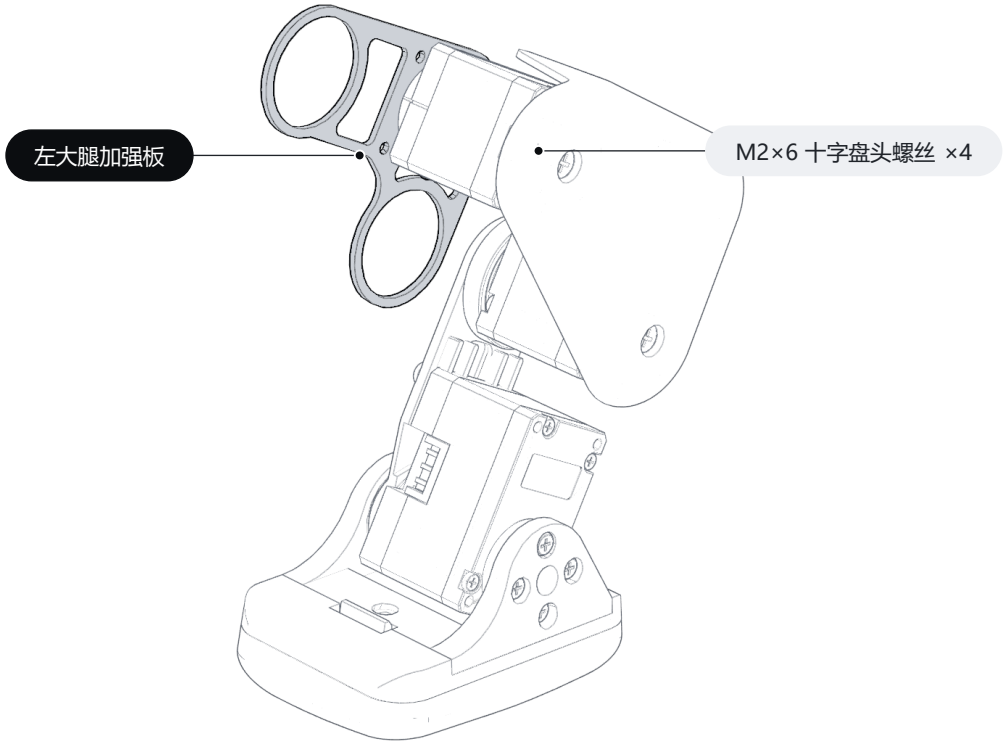
舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

L08 大腿加强板

THIGH STIFFENER PLATE





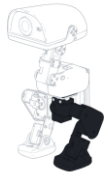
先松后紧 SNUG, THEN TIGHTEN

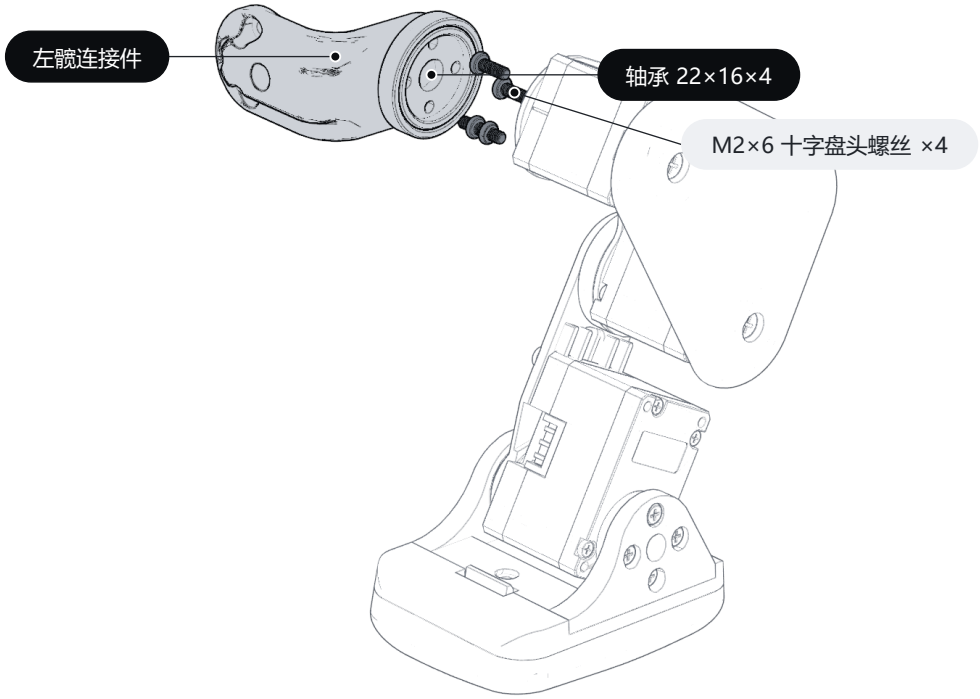
加强板螺丝先全部带上，再对角拧紧，避免大腿扭曲。

Start every stiffener screw, then tighten crosswise so the thigh is not twisted.

L09 左髋连接件

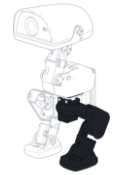
LEFT HIP BRACKET

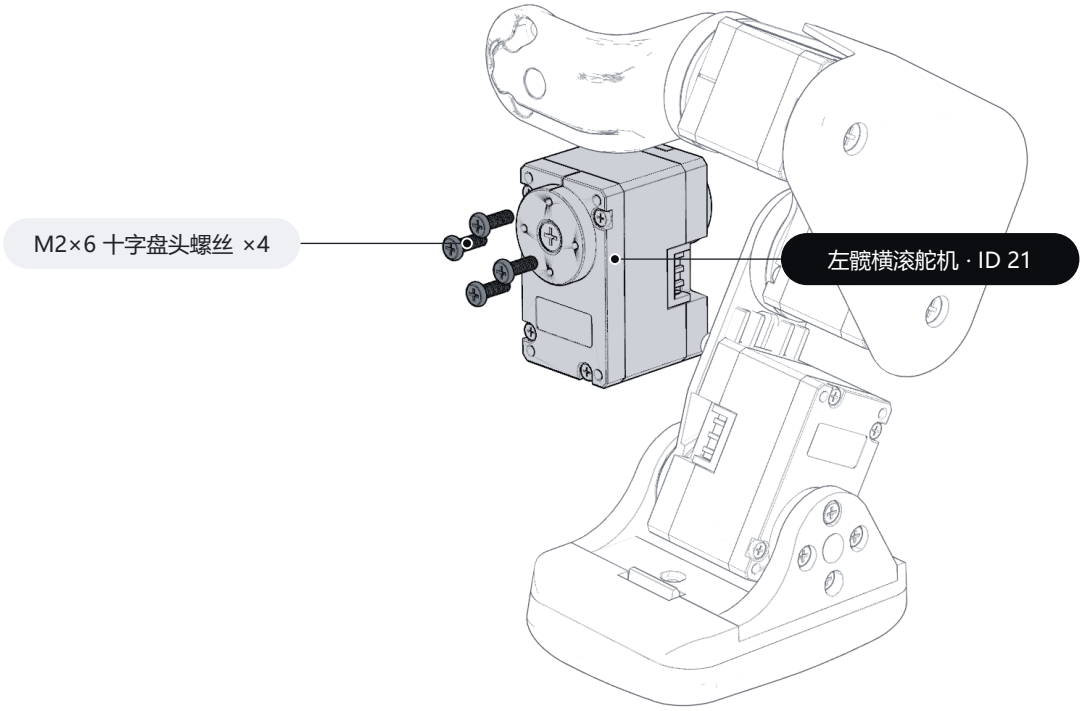




L10 左髋横滚舵机

LEFT HIP ROLL SERVO





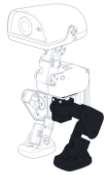
先对零位 ZERO FIRST

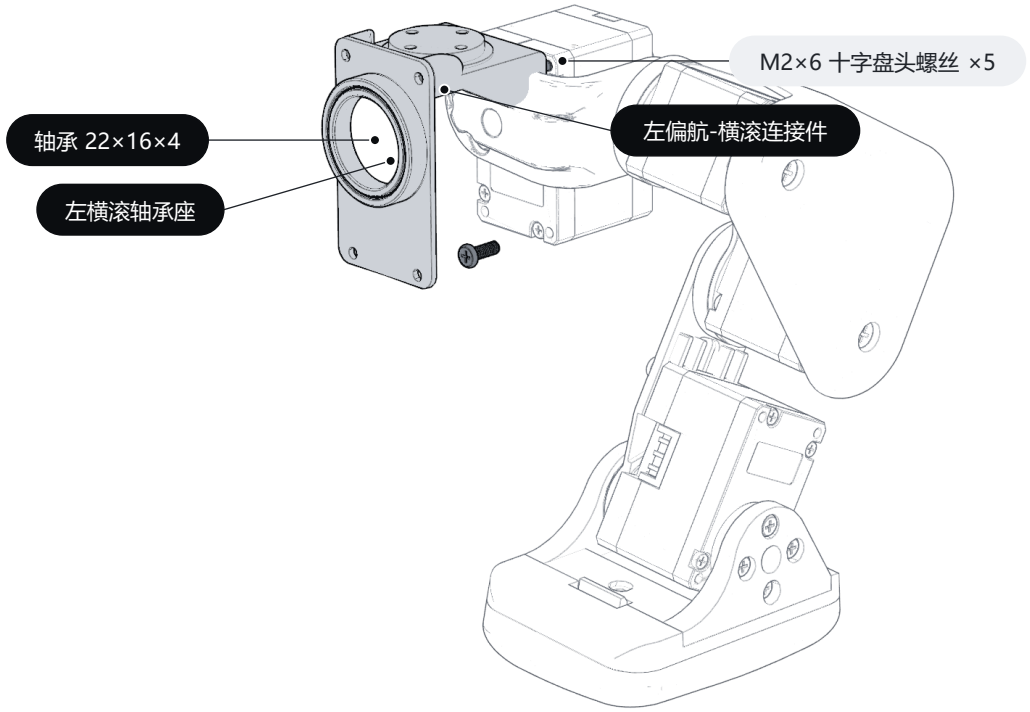
舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

L11 偏航-横滚连接件

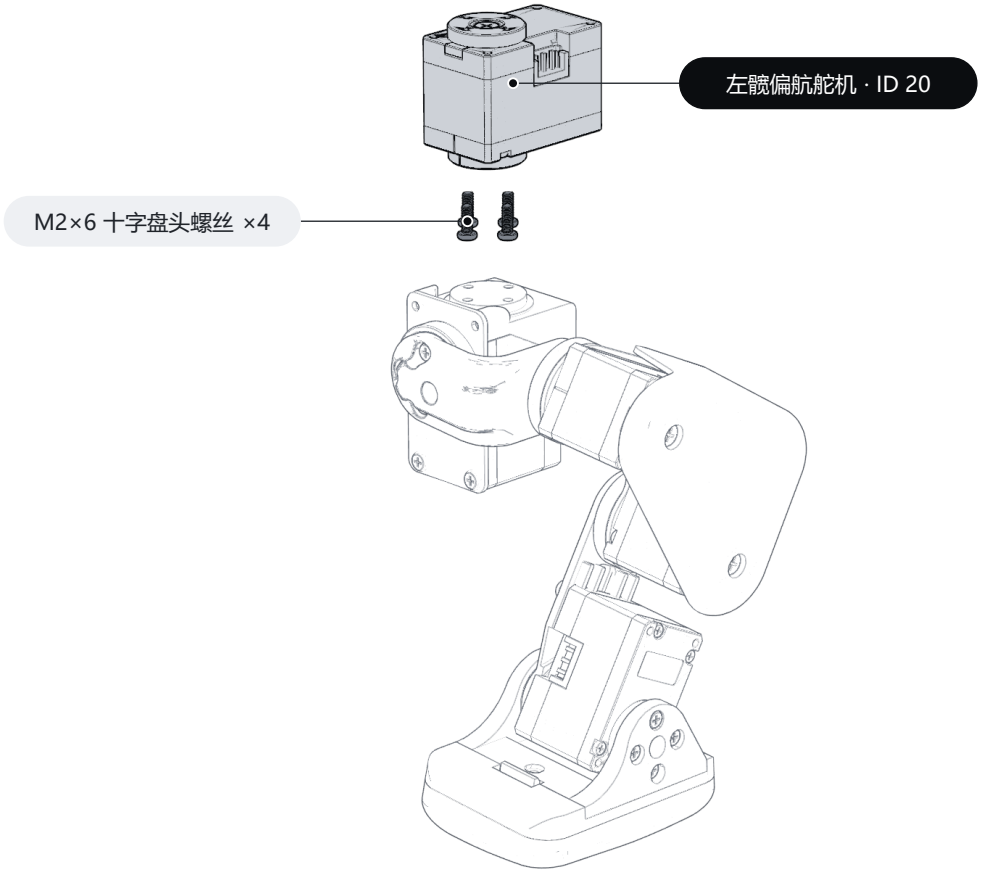
YAW-TO-ROLL BRACKET





L12 左髋偏航舵机

LEFT HIP YAW SERVO





先对零位 ZERO FIRST

舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

整腿检查 LEG CHECK

五个关节都能在支撑下小幅转动，线缆出口朝向躯干一侧，不被连杆压住。

All five joints move a little by hand under support; cable exits face the trunk and are not pinched.

02

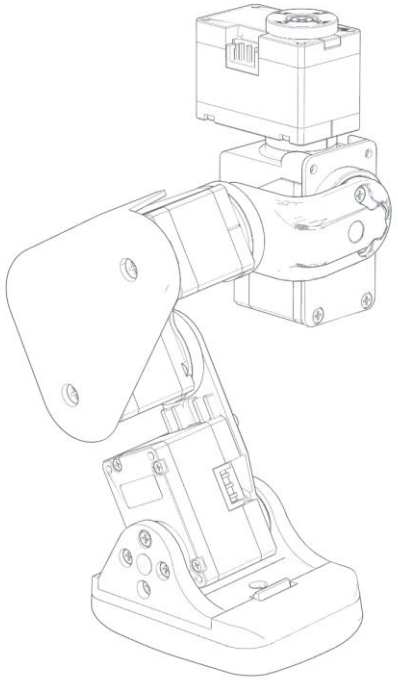
右腿

RIGHT LEG

12 步 · R01–R12

右脚底 Right sole	**×1**
右脚掌 Right foot	**×1**
右踝座 Right ankle bracket	**×1**
小轴承 Small bearing	**×1**
右踝舵机 · ID 14 R ankle servo · ID 14	**×1**
右小腿 Right lower leg	**×1**
右膝舵机 · ID 13 R knee servo · ID 13	**×1**
右大腿 Right thigh	**×1**
轴承 22×16×4 Bearing 22×16×4	**×3**
右髋俯仰舵机 · ID 12 R hip pitch servo · ID 12	**×1**
右大腿加强板 Right thigh stiffener	**×1**
右髋连接件 Right hip bracket	**×1**
右髋横滚舵机 · ID 11 R hip roll servo · ID 11	**×1**
右偏航-横滚连接件 Right yaw-to-roll bracket	**×1**
右横滚轴承座 Right roll bearing housing	**×1**
右髋偏航舵机 · ID 10 R hip yaw servo · ID 10	**×1**
M2×6 十字盘头螺丝 M2×6 pan head screw	**×25**
M2×8 十字盘头螺丝 M2×8 pan head screw	**×6**





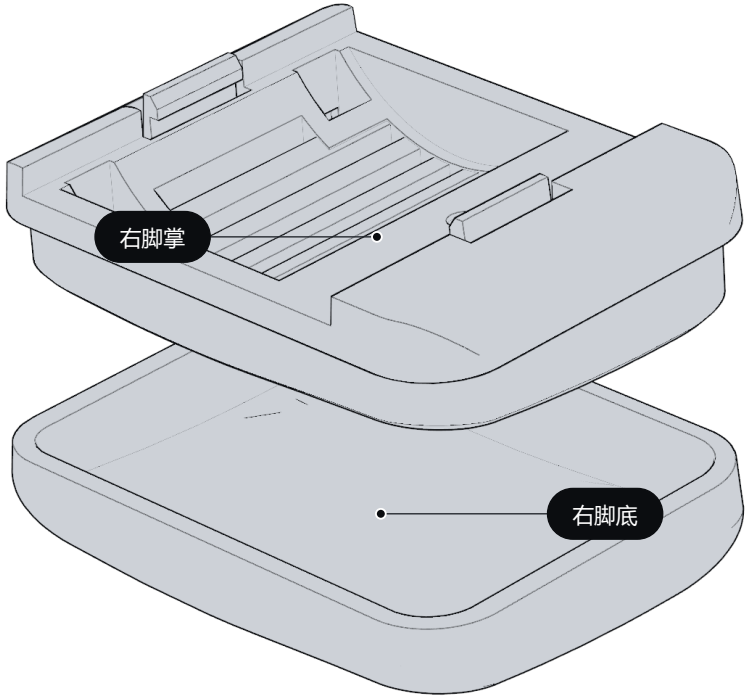
扫码体验 Microduck 可视化 Agent 服务

Scan to try the Microduck visual agent

R01 右脚掌与脚底

RIGHT FOOT AND SOLE





配合方式 FIT

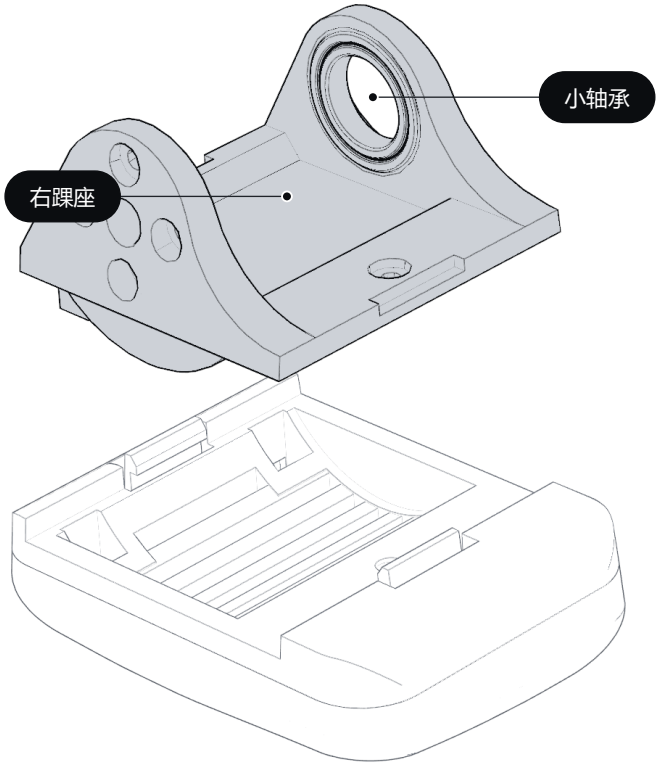
模型中脚底与脚掌之间没有紧固件，按实物卡槽/胶合方式装配。

No fastener is modelled between sole and foot; follow the physical snap or glue fit.

R02 右踝座与小轴承

RIGHT ANKLE BRACKET AND BEARING





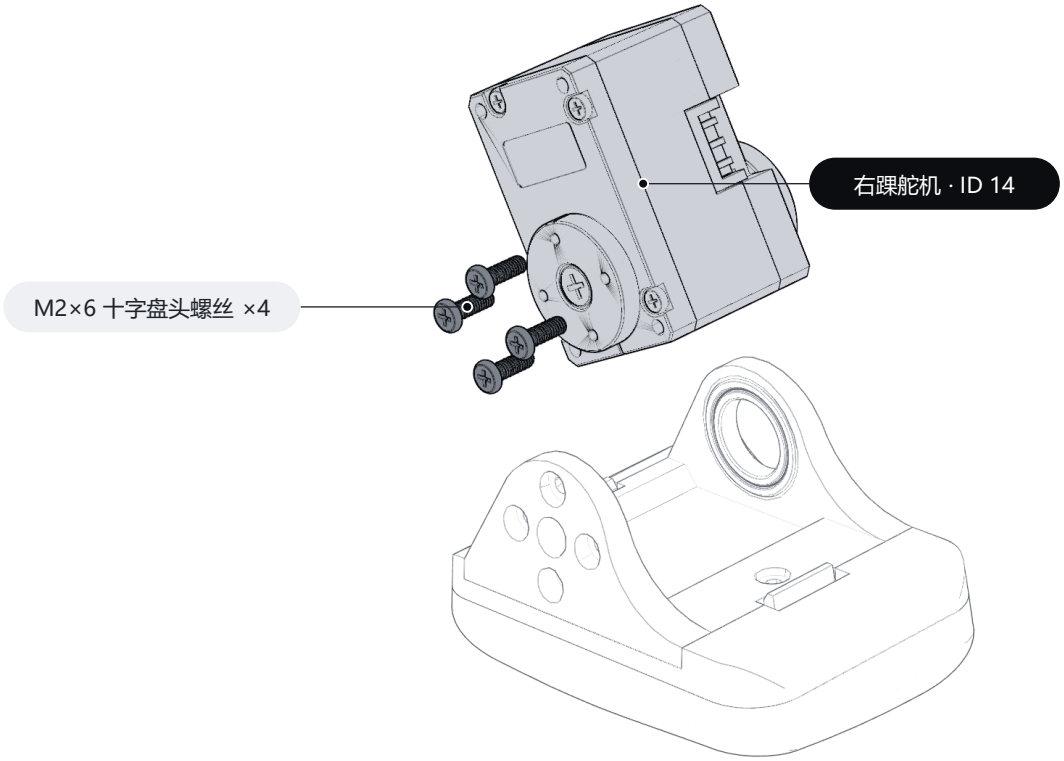
轴承到底 SEAT THE BEARING

小轴承压入踝座轴承孔，端面与孔口齐平。

Press the small bearing fully into the ankle bracket bore, flush with the face.

R03 右踝舵机
RIGHT ANKLE SERVO





先对零位 ZERO FIRST

舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

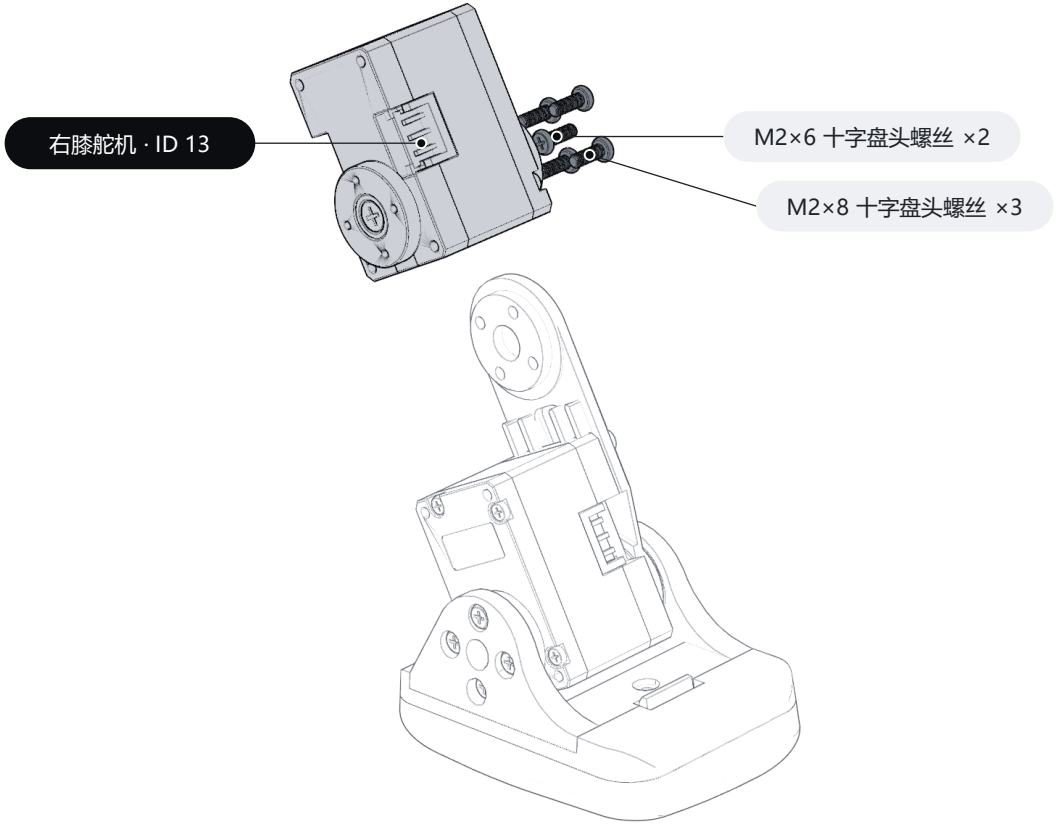
R04 右小腿
RIGHT LOWER LEG





R05 右膝舵机
RIGHT KNEE SERVO





先对零位 ZERO FIRST

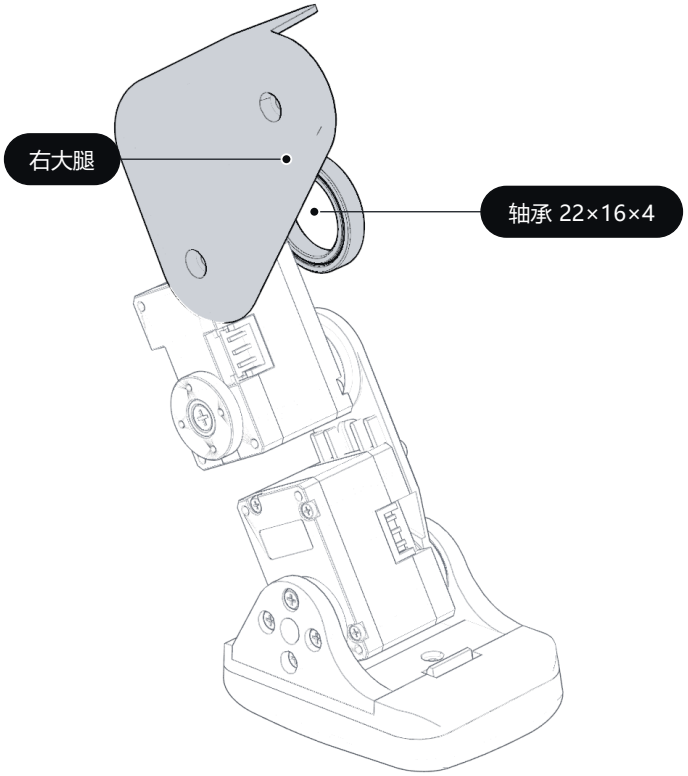
舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

R06 右大腿与轴承

RIGHT THIGH AND BEARING

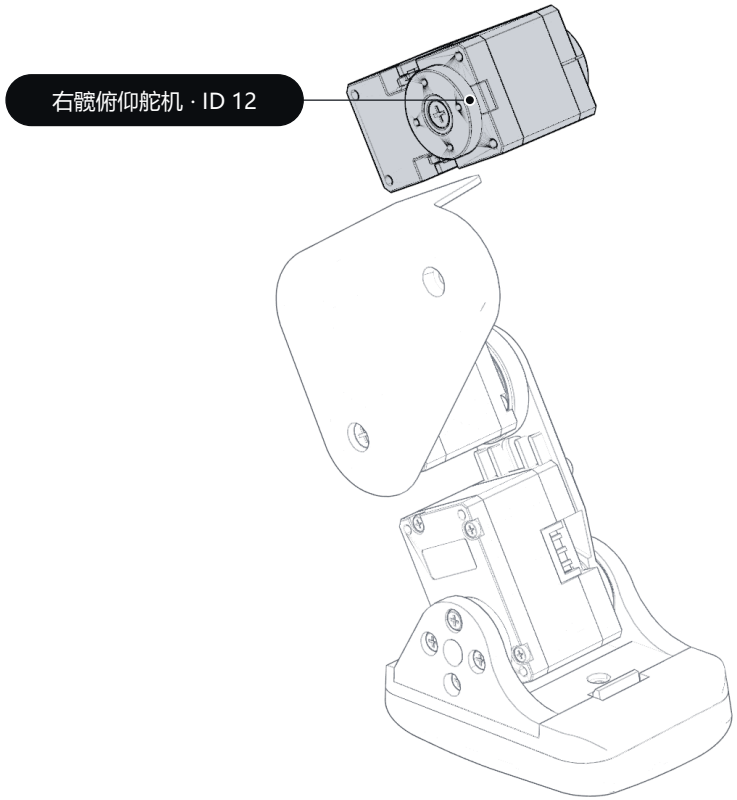




R07 右髋俯仰舵机

RIGHT HIP PITCH SERVO





先对零位 ZERO FIRST

舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

R08 大腿加强板

THIGH STIFFENER PLATE





先松后紧 SNUG, THEN TIGHTEN

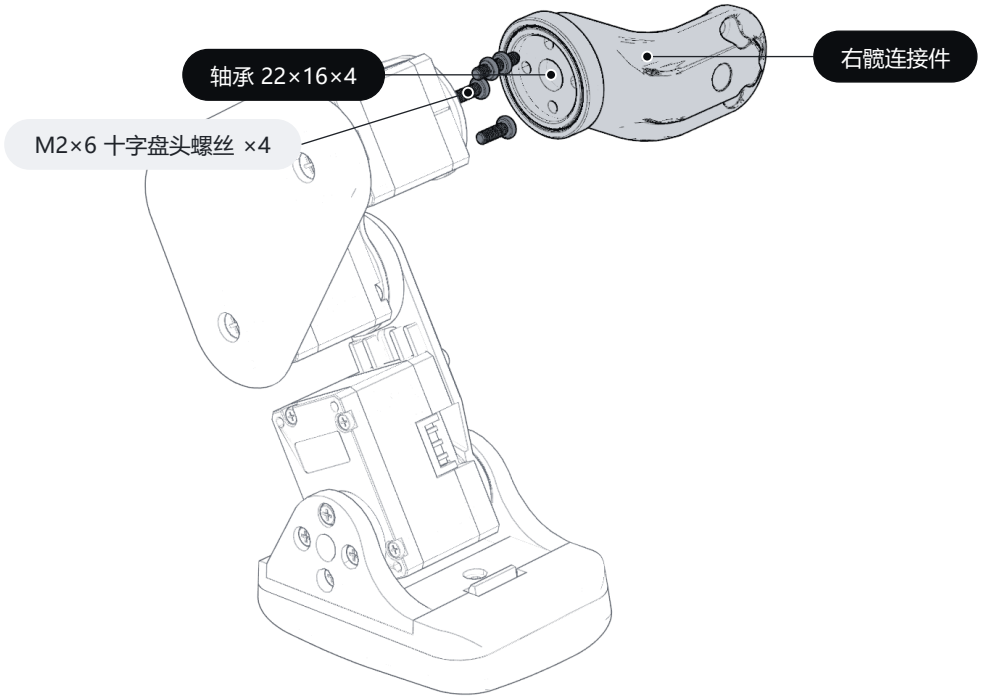
加强板螺丝先全部带上，再对角拧紧，避免大腿扭曲。

Start every stiffener screw, then tighten crosswise so the thigh is not twisted.

R09 右髋连接件

RIGHT HIP BRACKET

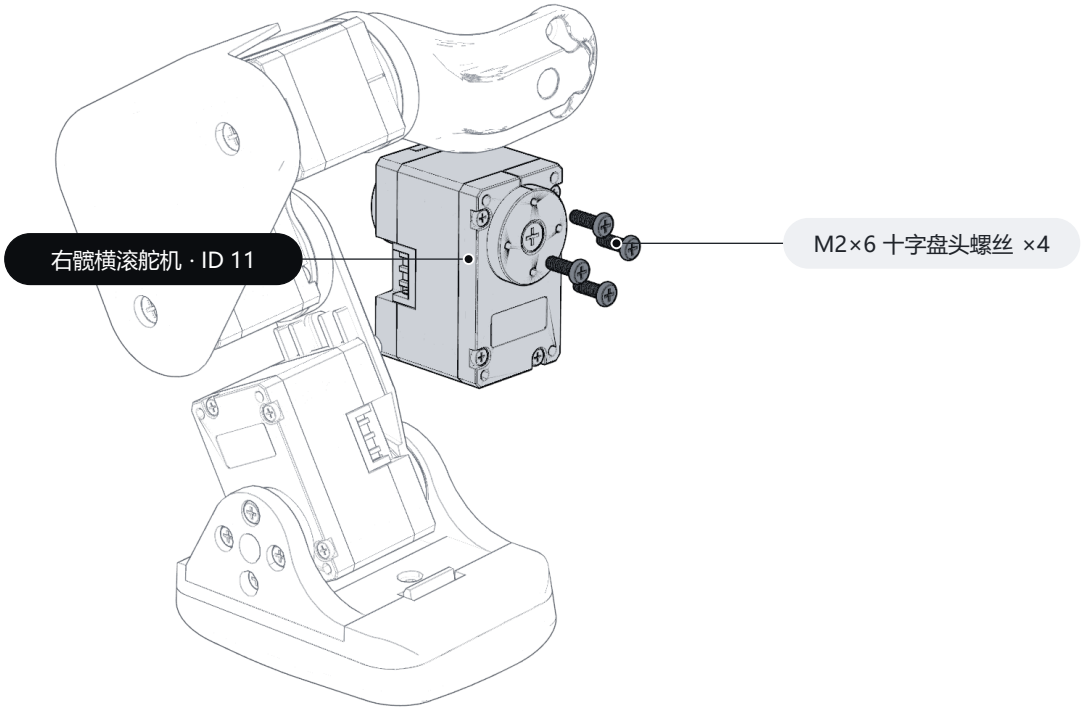




R10 右髋横滚舵机

RIGHT HIP ROLL SERVO





先对零位 ZERO FIRST

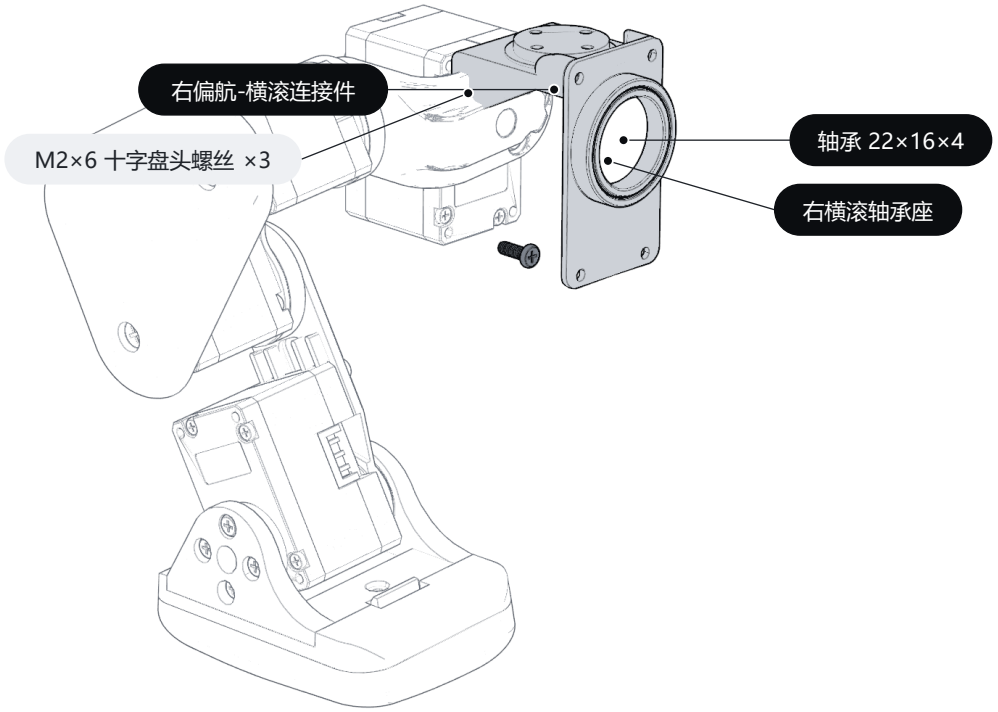
舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

R11 偏航-横滚连接件

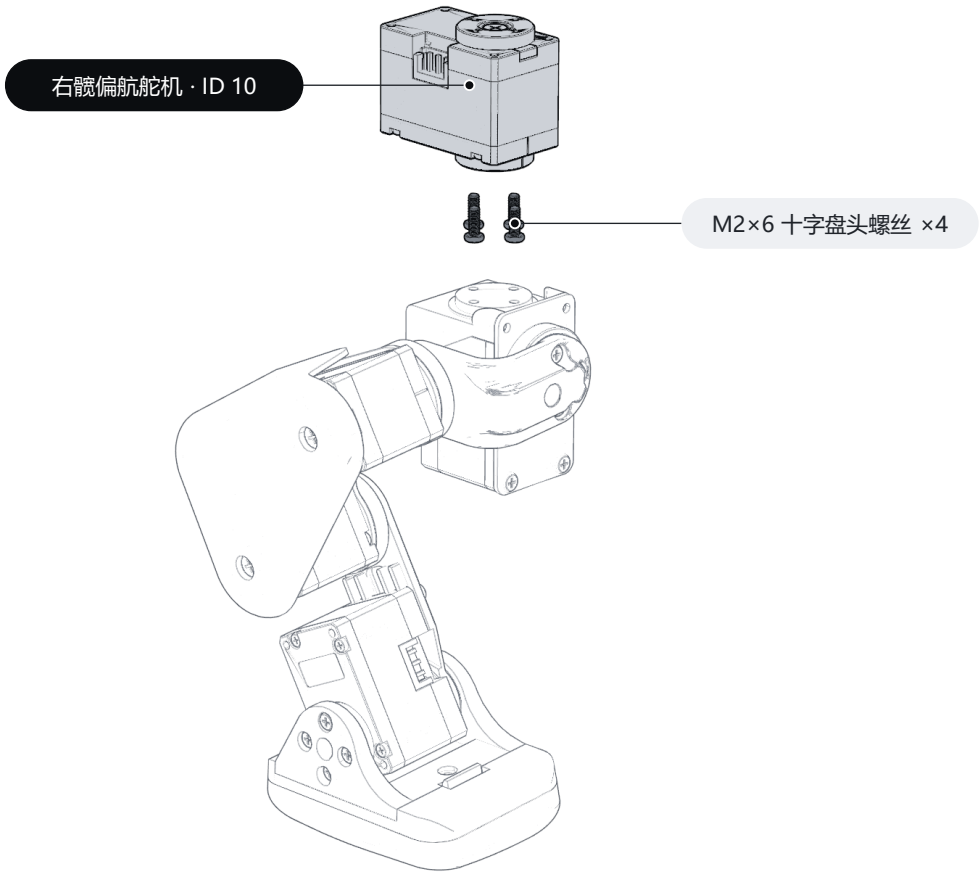
YAW-TO-ROLL BRACKET





R12 右髋偏航舵机

RIGHT HIP YAW SERVO





先对零位 ZERO FIRST

舵机装入前按软件手册 S1.4 配好 ID，并让舵盘处于中位（2048）再装连杆，避免错齿。

Set the ID (software manual S1.4) and park the horn at centre (2048) before fitting the link to avoid a tooth offset.

整腿检查 LEG CHECK

五个关节都能在支撑下小幅转动，线缆出口朝向躯干一侧，不被连杆压住。

All five joints move a little by hand under support; cable exits face the trunk and are not pinched.

03

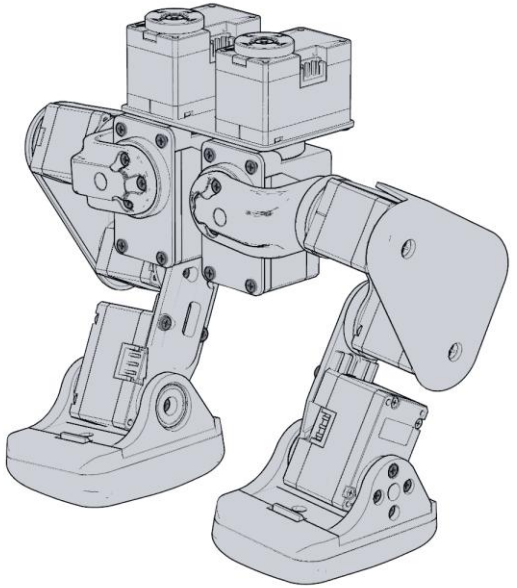


躯干与双腿合装

TRUNK & LEGS

3 步 · T01–T03

躯干底座 Trunk base	×1
轴承 22×16×4 Bearing 22×16×4	×2
M2×6 十字盘头螺丝 M2×6 pan head screw	×8



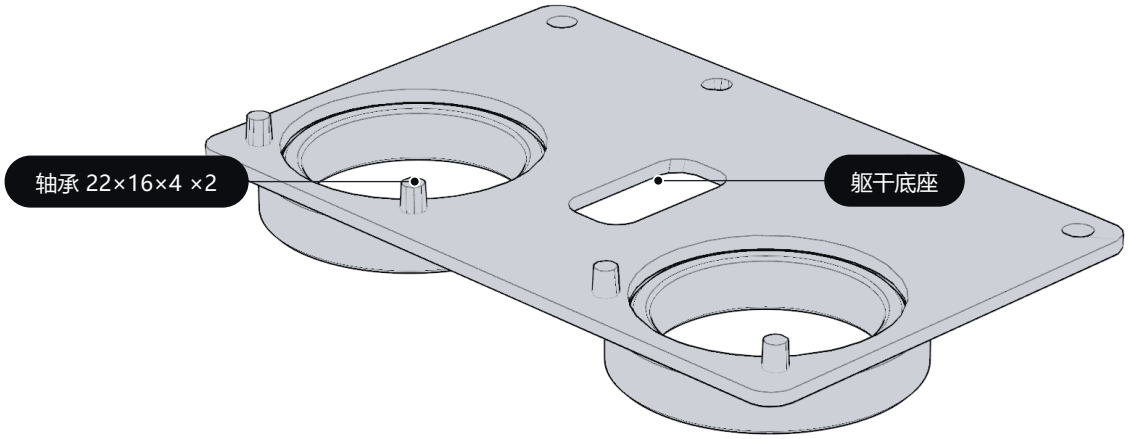
扫码体验 Microduck 可视化 Agent 服务

Scan to try the Microduck visual agent

T01 躯干底座与轴承

TRUNK BASE AND BEARINGS

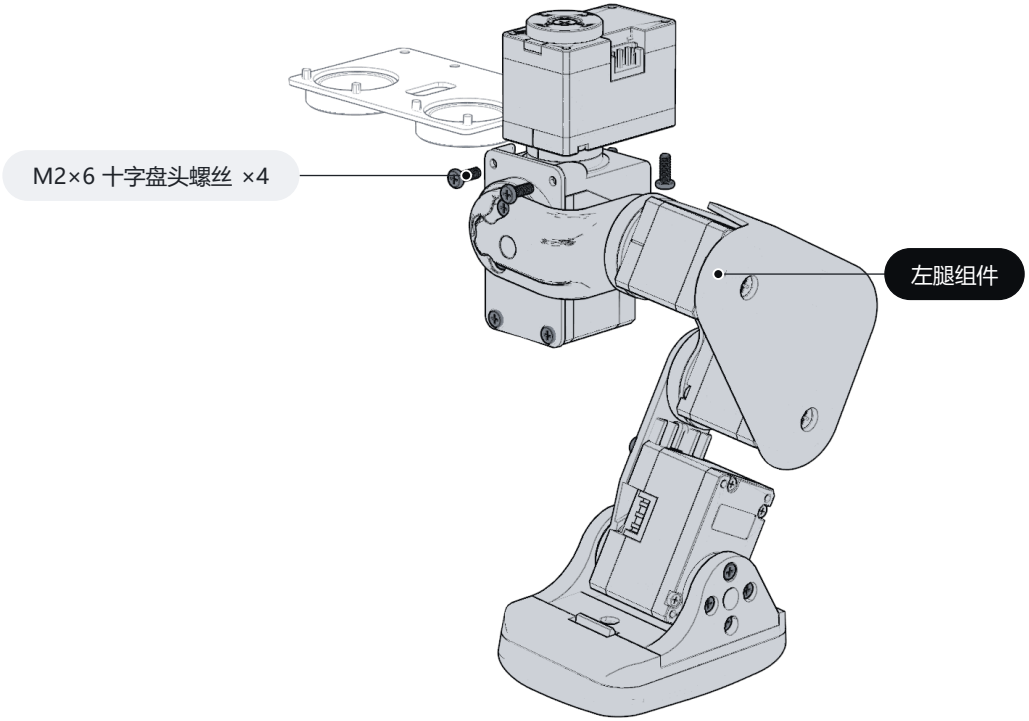




T02 装入左腿

FIT THE LEFT LEG





方向 ORIENTATION

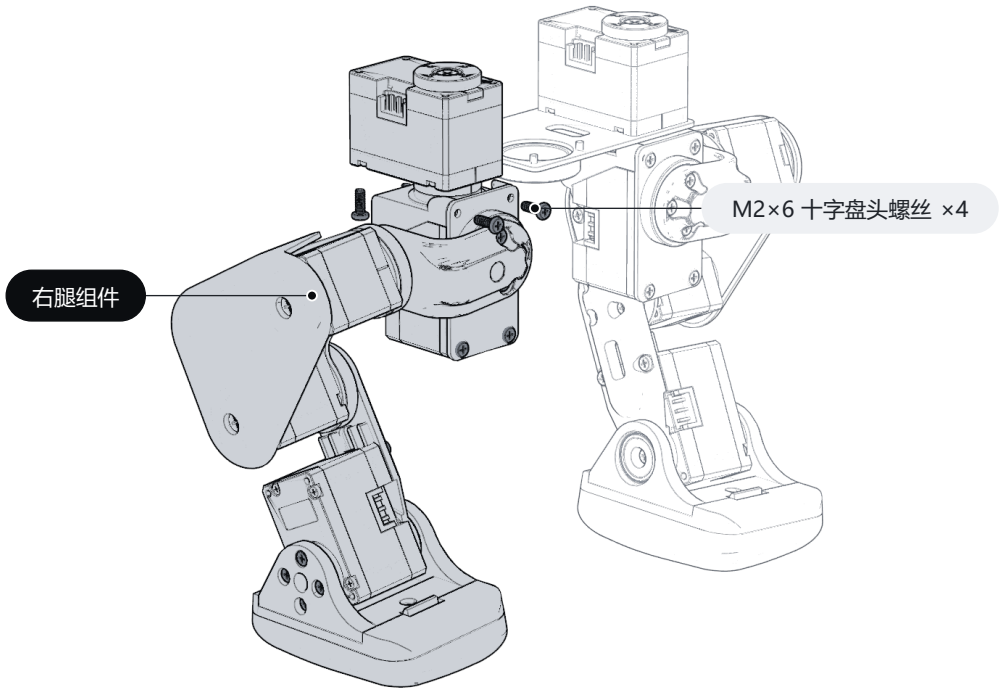
左腿以机器人自身为准：面朝前方时在左侧（图中 −Z 侧）。

Left is the robot's own left when it faces forward.

T03 装入右腿

FIT THE RIGHT LEG







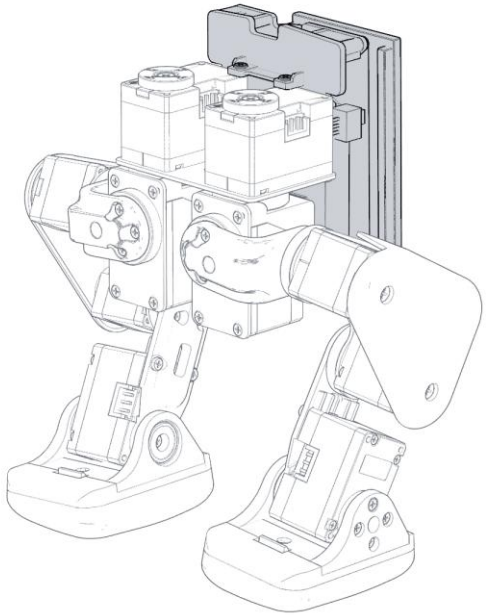
04

电源与电池

POWER & BATTERY

5 步 · E01–E05

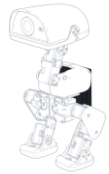
电源支架 Power support	×1
IMU 总线板 imu_to_dxl IMU bus board imu_to_dxl	×1
电池座 Battery sled	×1
电源开关 Power switch	×1
PCB 卡扣 PCB locker	×1
NP-F550 电池 NP-F550 battery	×1
M2×6 十字盘头螺丝 M2×6 pan head screw	×12

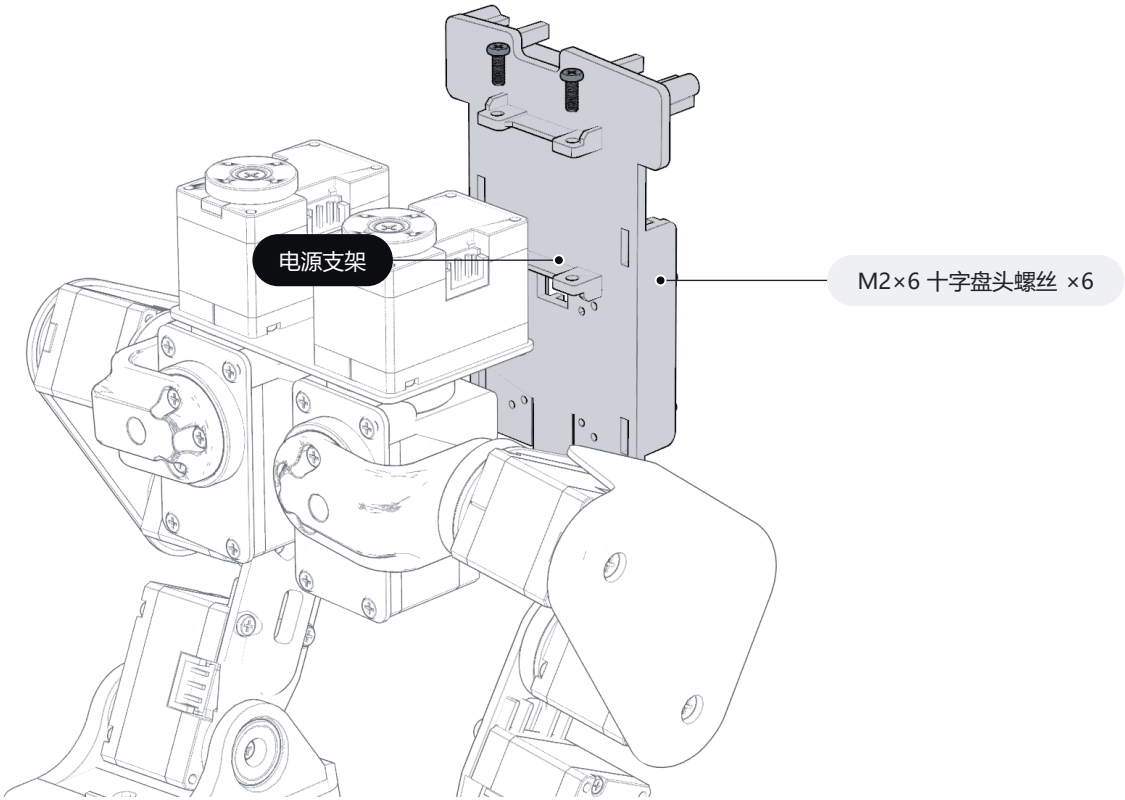


扫码体验 Microduck 可视化 Agent 服务

Scan to try the Microduck visual agent

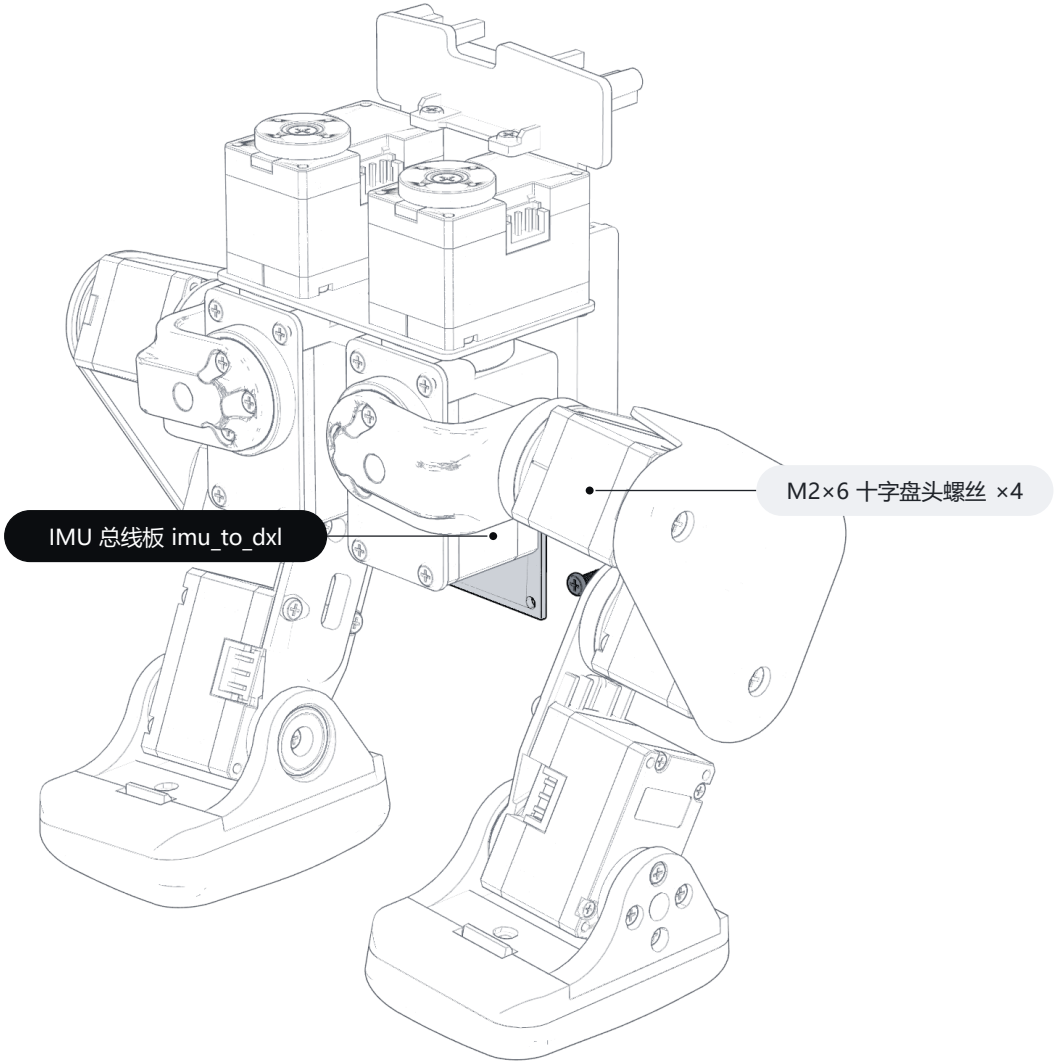
E01 电源支架
POWER SUPPORT

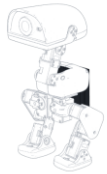




E02 IMU 总线板

IMU BUS BOARD





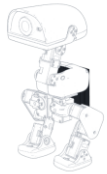
刚性固定 MOUNT RIGIDLY

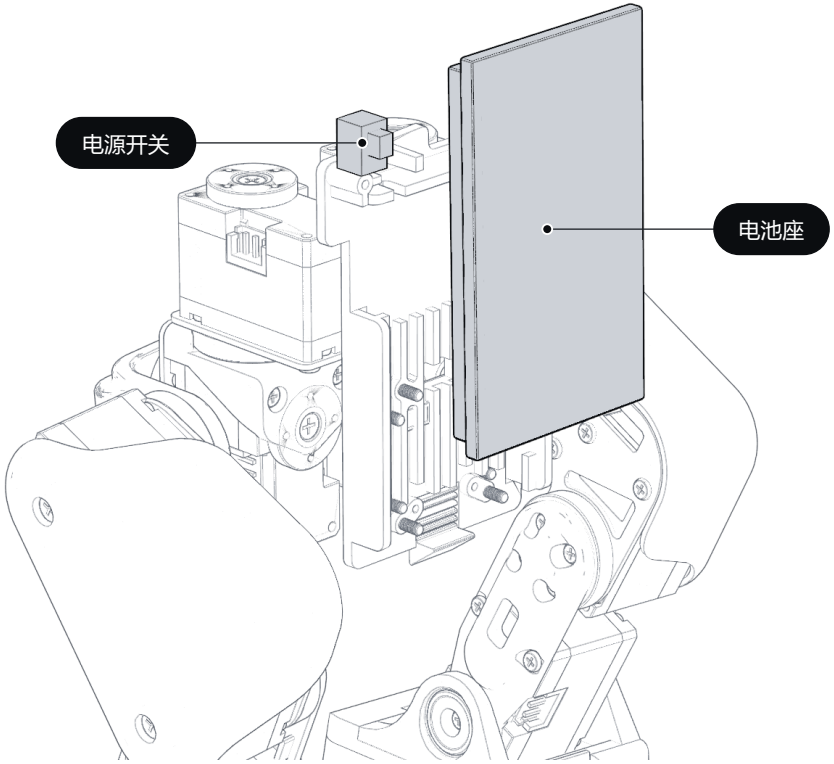
IMU 板必须刚性固定在躯干上，安装方向与软件手册 S9.3 核对。

Mount the IMU board rigidly; verify its orientation with software manual S9.3.

E03 电池座与电源开关

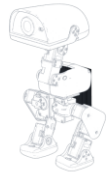
BATTERY SLED AND SWITCH

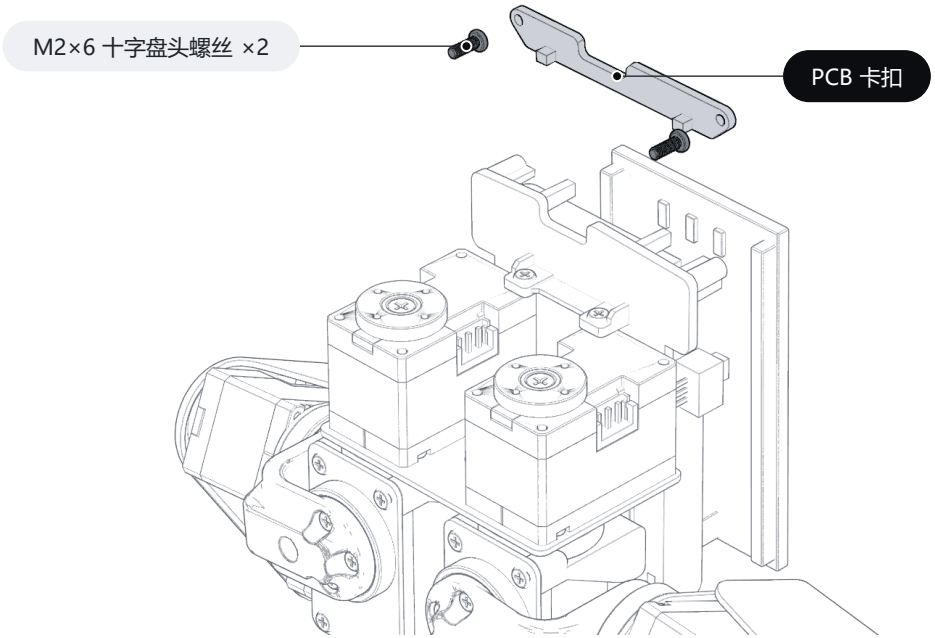




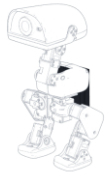
E04 PCB 卡扣

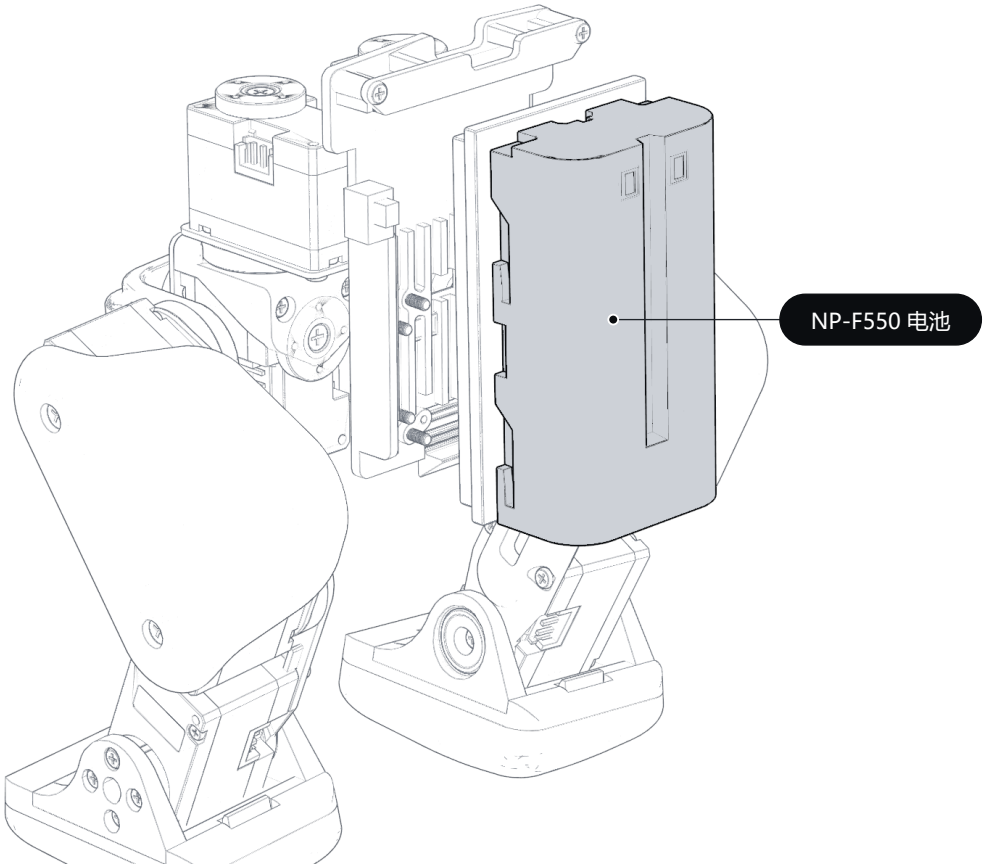
PCB LOCKER





E05 电池
BATTERY





供电冲突 VOLTAGE CONFLICT

零售 XL330 额定 3.7–6.0 V，而 2S 电池满电约 8.2 V。供电方案闭合前不要用电池给整机带载（软件手册 S9.1）。

Retail XL330 is rated 3.7–6.0 V but a 2S pack reaches about 8.2 V. Do not load the robot from the battery until the power plan is closed (software manual S9.1).

05

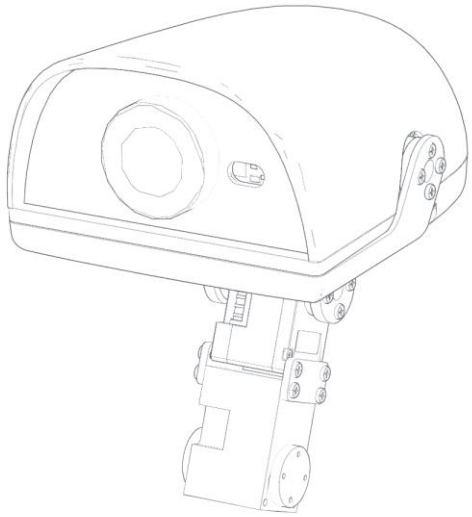
颈部与头部组件

NECK & HEAD ASSEMBLY

12 步 · H01–H12

颈座 Neck bracket	×2
颈部舵机 A Neck servo A	×1
颈部舵机 B Neck servo B	×1
颈俯仰件 Neck pitch link	×1
轴承 22×16×4 Bearing 22×16×4	×3
头部偏航-横滚件 Head yaw-roll link	×1
头部偏航-横滚舵机 Head yaw-roll servo	×1
头部电机支架 Head motor support	×1
头部舵机 Head servo	×1
鸭嘴舵机 Beak servo	×1
下颌 Jaw	×1
软下颌 Soft jaw	×1
小轴承 Small bearing	×1
头壳（下） Bottom head shell	×1
面板 Face plate	×1
M12 镜头座 M12 lens holder	×1
镜头 Lens	×1
眼部饰件 Eye trim	×1
ToF 深度传感器 ToF sensor	×1
Radxa Zero 3W 主控 Radxa Zero 3W	×1
microSD 卡 microSD card	×1
Robot HAT 载板 Robot HAT	×1
扬声器 Speaker	×1
NFC 天线 · 头部 NFC antenna · head	×1
NFC 天线 · 鸭嘴 NFC antenna · beak	×1
软上喙 Soft upper beak	×1
头壳（上） Top head shell	×1
M2 十字盘头螺丝 M2 pan head screw	×8
M2×4 十字盘头螺丝 M2×4 pan head screw	×2
M2×6 十字盘头螺丝 M2×6 pan head screw	×27
M2×8 十字盘头螺丝 M2×8 pan head screw	×11



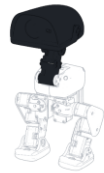


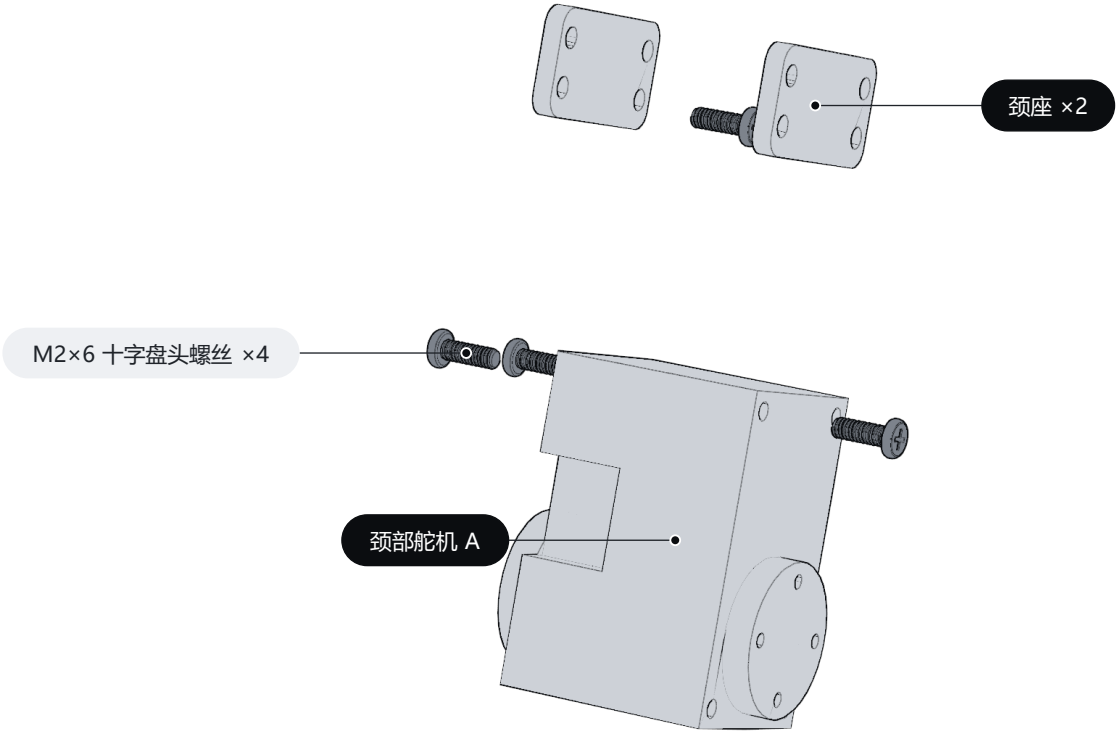
扫码体验 Microduck 可视化 Agent 服务

Scan to try the Microduck visual agent

H01 颈座与颈部舵机 A

NECK BRACKET AND NECK SERVO A





头部舵机 ID HEAD SERVO IDS

头部五颗舵机的 ID 按 30–34（颈俯仰、头俯仰、头偏航、头横滚、嘴）配置；本手册只按位置标注，装机后按软件手册 S9.2 逐个活动核对。

Head servos take IDs 30–34 (neck pitch, head pitch, head yaw, head roll, mouth). They are labelled here by position only; confirm each by moving it (software manual S9.2).

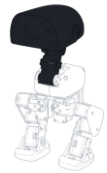
舵盘朝外 HORN OUTWARD

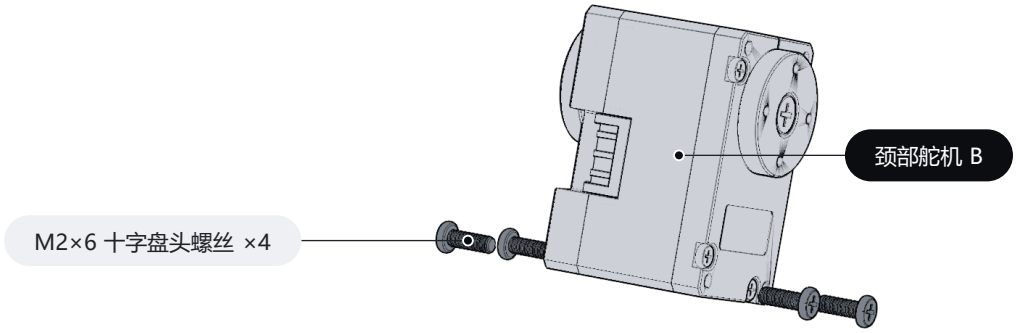
颈部舵机 A 之后要与左右侧壳相连，两侧都要留出螺丝通道。

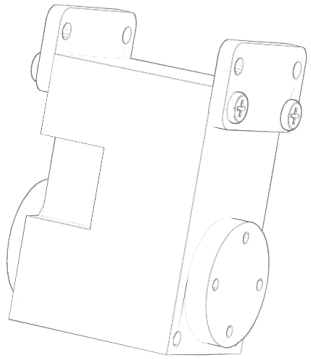
Neck servo A later bolts to both side shells; keep both faces accessible.

H02 颈部舵机 B

NECK SERVO B

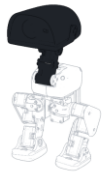


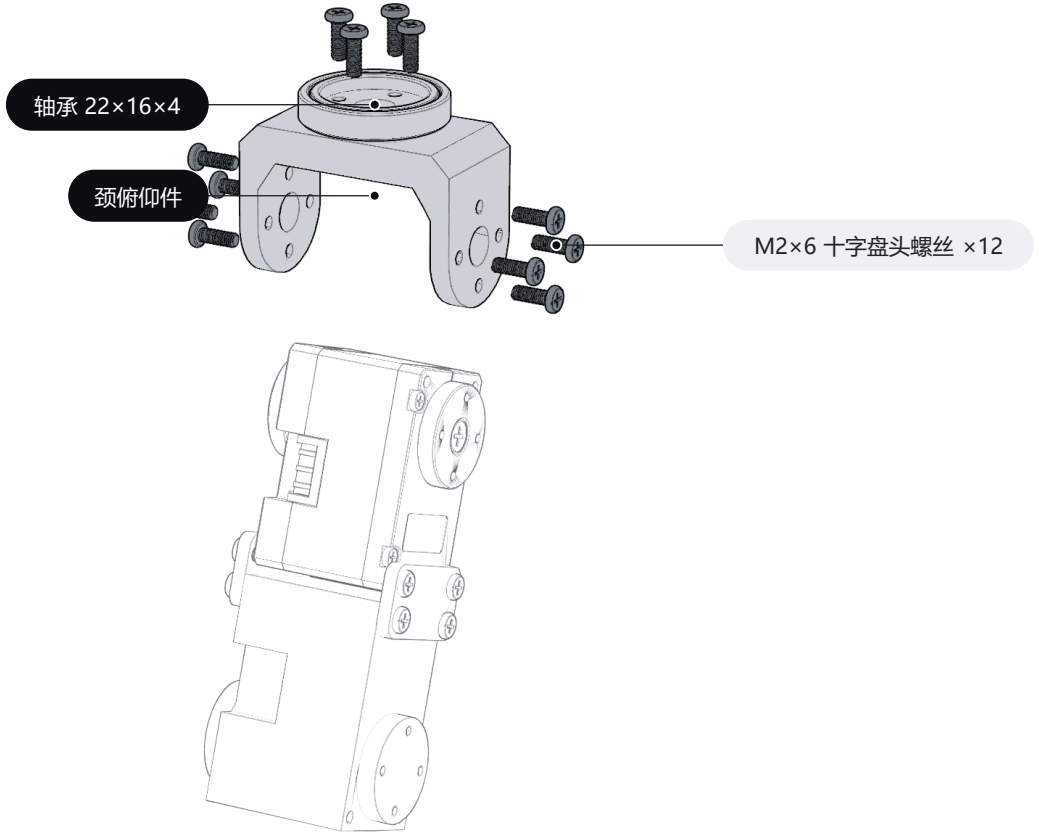




H03 颈俯仰件

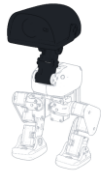
NECK PITCH LINK

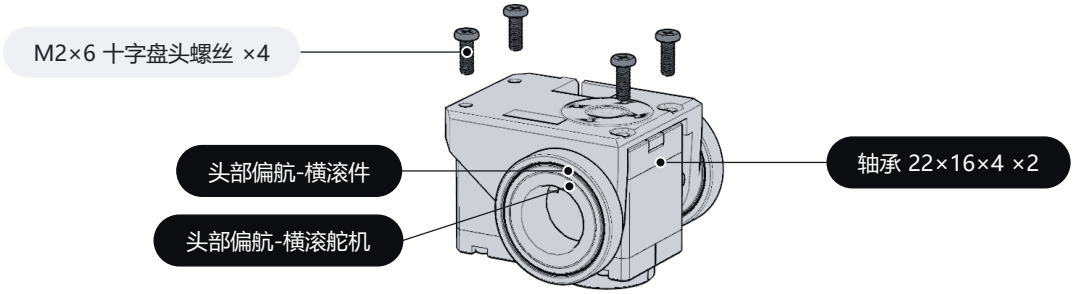


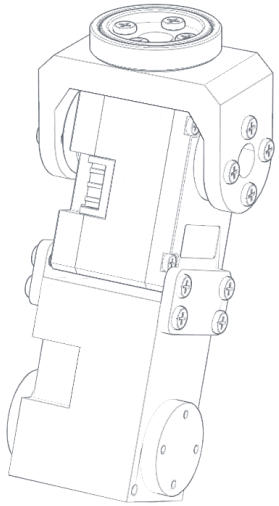


H04 头部偏航-横滚件与舵机

HEAD YAW-ROLL LINK AND SERVO

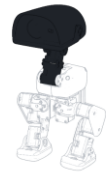


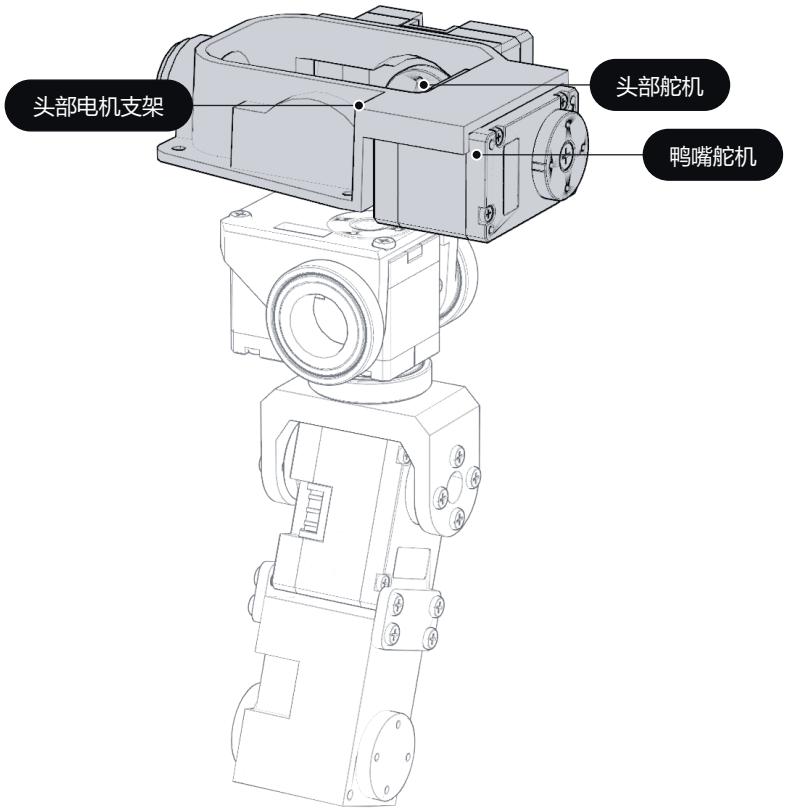




H05 头部电机支架与舵机

HEAD MOTOR SUPPORT AND SERVOS



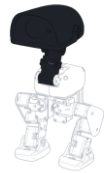


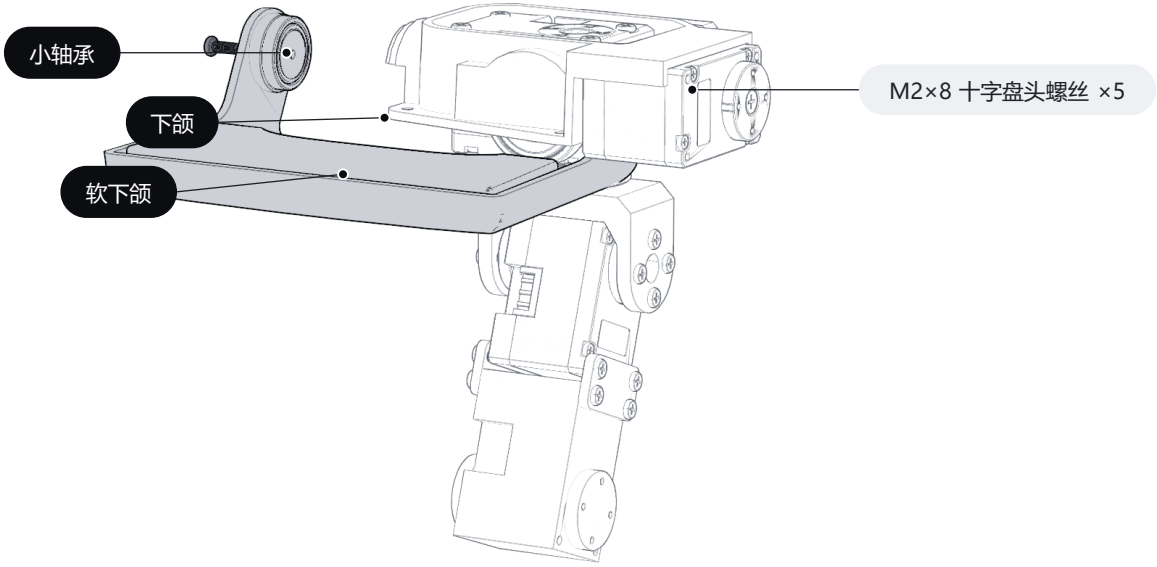
两颗舵机 TWO SERVOS

头部舵机装在支架后部，鸭嘴舵机装在侧面，舵盘朝向下颌一侧。

The head servo sits at the rear of the support and the beak servo on the side, horn towards the jaw.

H06 下颌
JAW



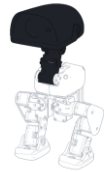


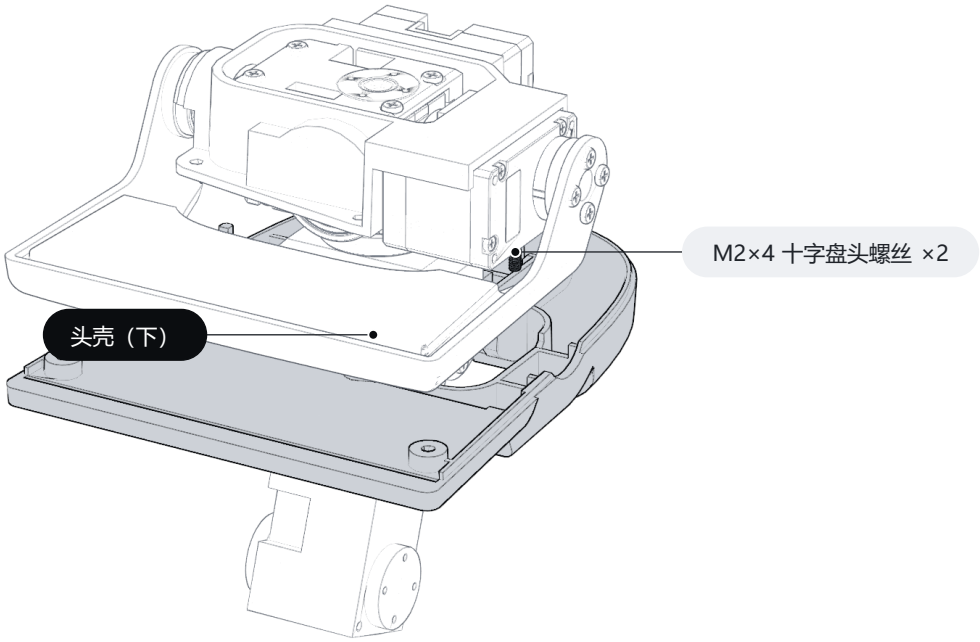
先试转 CHECK TRAVEL

下颌装上后先手动转动，确认能完全合拢、不碰头壳。

Swing the jaw by hand to confirm it closes fully without touching the shell.

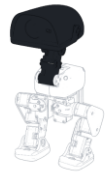
H07 头壳（下）
BOTTOM HEAD SHELL

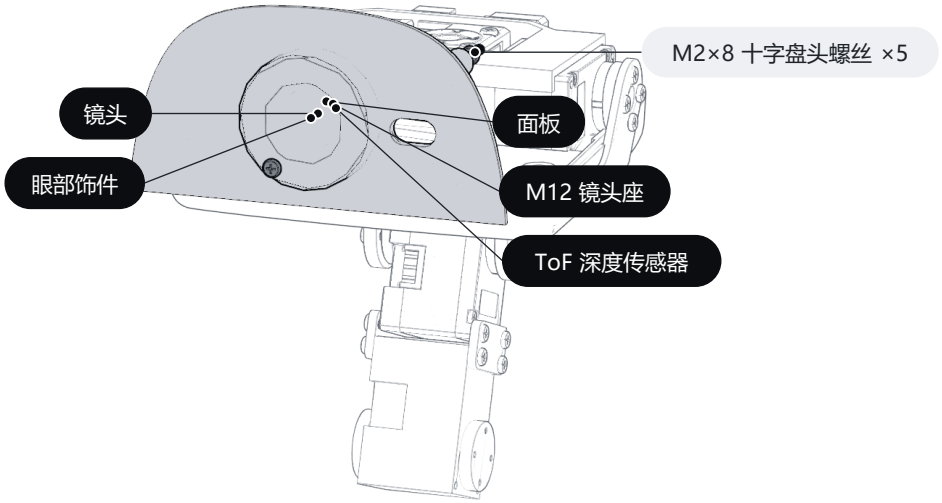




H08 面板、镜头与 ToF

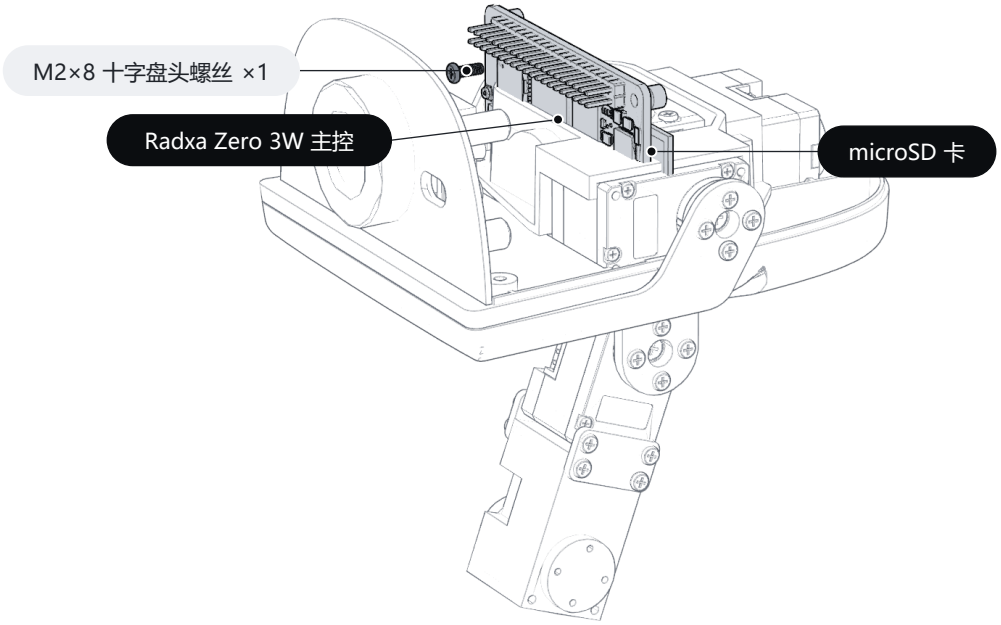
FACE PLATE, LENS AND TOF

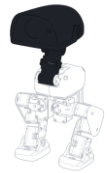




H09 Radxa Zero 3W 与 microSD

RADXA ZERO 3W AND MICROSD





先写卡 FLASH FIRST

microSD 先按软件手册 S3 写好系统再插入。

Flash the microSD card (software manual S3) before inserting it.

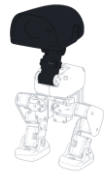
主控在头部 BOARD IN THE HEAD

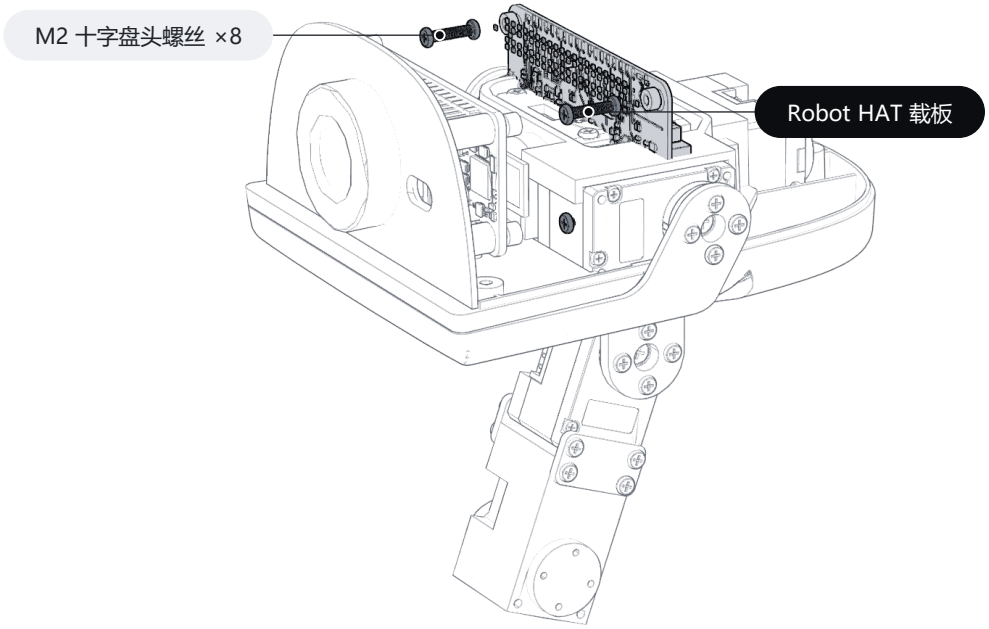
该机型的主控竖直装在面板后方。

On this build the main board stands upright behind the face plate.

H10 Robot HAT 载板

ROBOT HAT





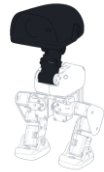
排针对位 HEADER ALIGNMENT

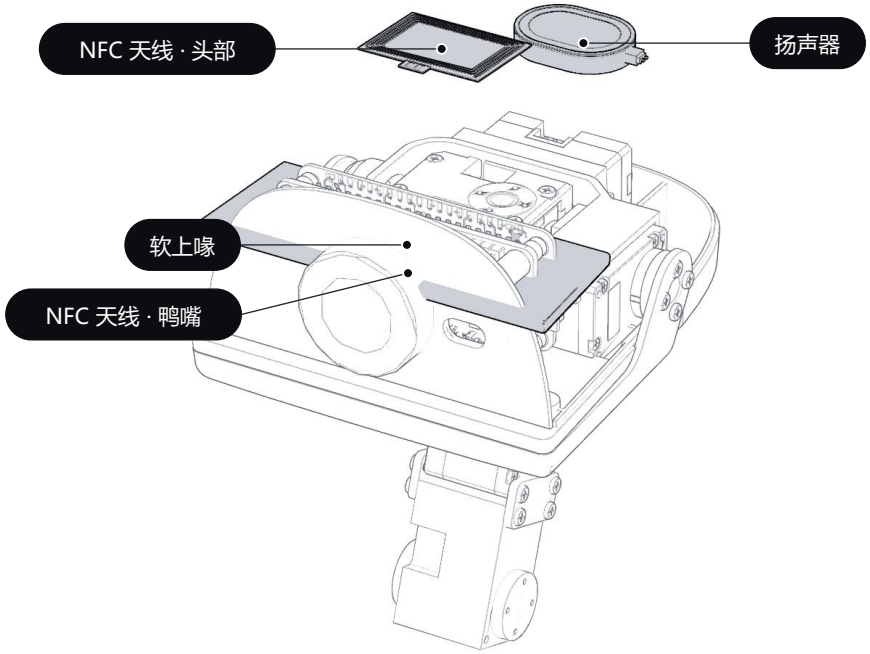
确认 2×20 排针 Pin 1 方向后再压入；插拔一律断电。

Check Pin 1 of the 2x20 header before pressing on; always power off to plug or unplug.

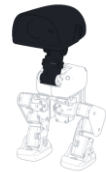
H11 扬声器、NFC 与软上喙

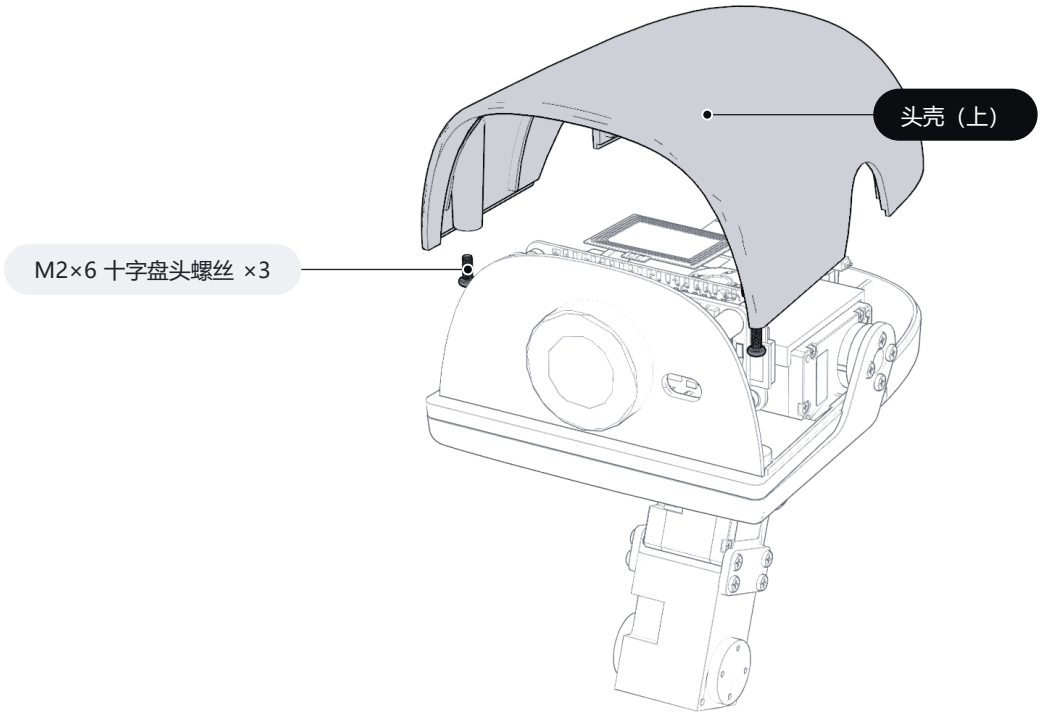
SPEAKER, NFC AND SOFT BEAK





H12 头壳（上）
TOP HEAD SHELL





走线再合壳 ROUTE, THEN CLOSE

合上头壳前确认总线与相机排线已穿过颈部，不被壳体压住。

Before closing, run the bus and camera cables down through the neck so the shell does not pinch them.



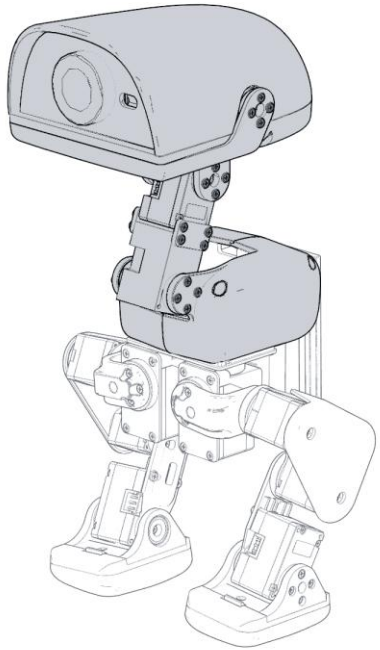
06

侧壳与整机合装

SHELLS & FINAL ASSEMBLY

3 步 · S01–S03

左侧壳 Left shell	×1
右侧壳 Right shell	×1
M2×6 十字盘头螺丝 M2×6 pan head screw	×2
M2×8 十字盘头螺丝 M2×8 pan head screw	×9

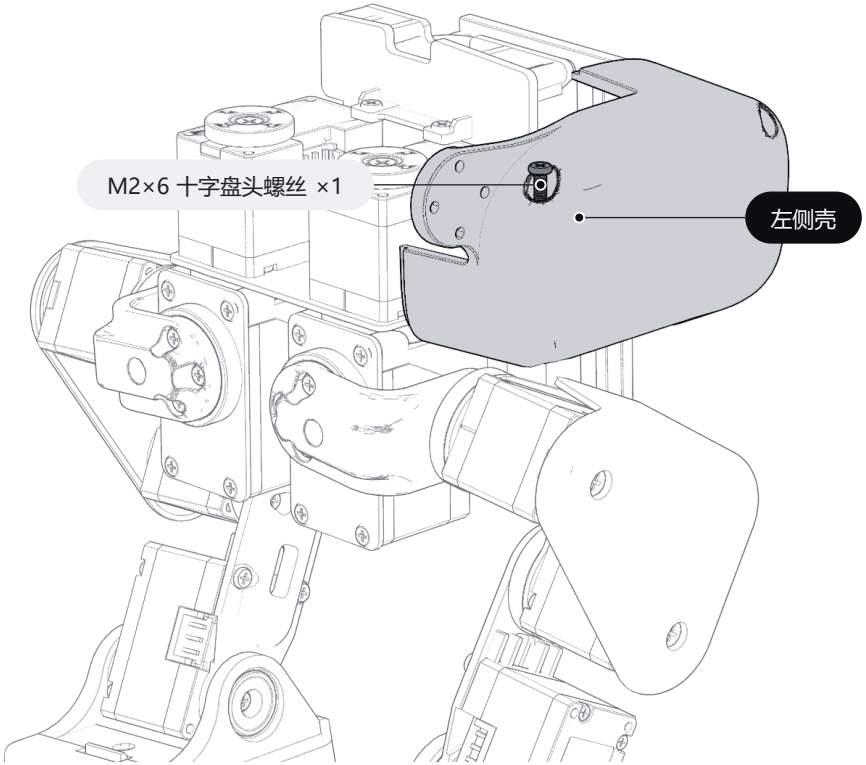


扫码体验 Microduck 可视化 Agent 服务

Scan to try the Microduck visual agent

S01 左侧壳
LEFT SHELL

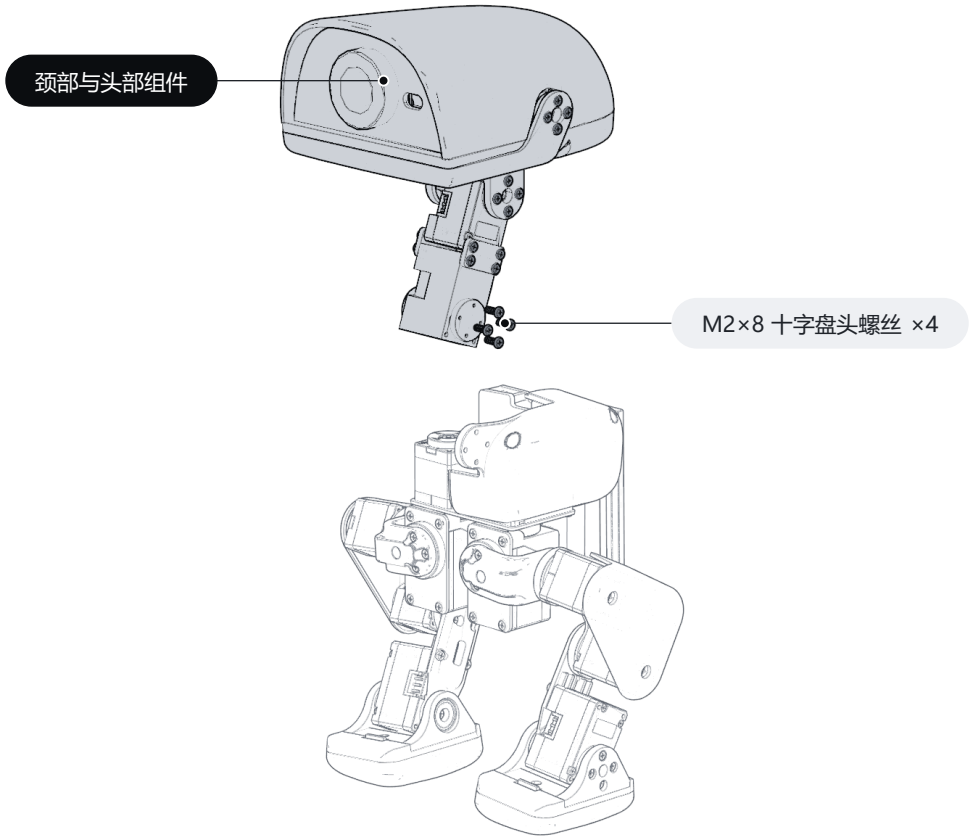




S02 装入颈部与头部

FIT THE NECK AND HEAD



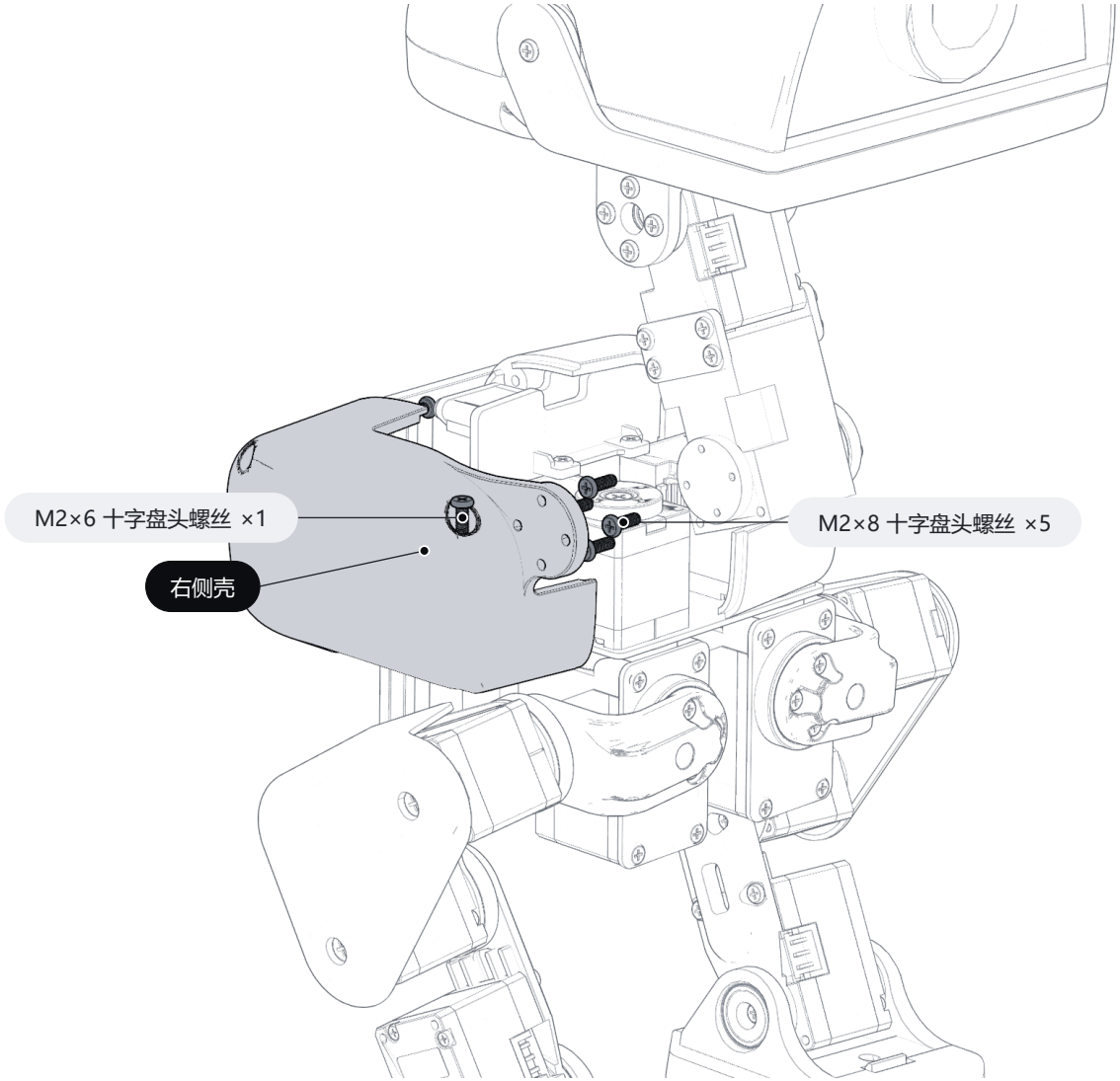


先左后右 LEFT, THEN RIGHT

颈部舵机 A 先与左侧壳固定，再装右侧壳把它夹住；两侧螺丝都带上后再逐颗拧紧。

Fix neck servo A to the left shell first, then close it with the right shell; start every screw on both sides before tightening.

S03 右侧壳
RIGHT SHELL





完工检查 FINAL CHECK

所有关节在支撑下能小幅转动、无夹线；进入软件手册 S8.2 整机上线。

Every joint moves a little by hand with no pinched cable; continue with software manual S8.2.

完成

COMPLETE

机械装配到此结束。接下来按《MicroDuck 装配与调试手册 V1.2》第 10–11 章：整机上线（S8.2）、上电前安全闸门（S9.1）、HOME 数字孪生比对（S9.2）与 IMU 方向核对（S9.3），再站立和行走。

Mechanical assembly is complete. Continue with the software manual V1.2: bus bring-up (S8.2), safety gate (S9.1), HOME check (S9.2) and IMU orientation (S9.3) before standing and walking.

扫码加入
Nowpop交流群

